PONDICHERRY UNIVERSITY
RAMANUJAN SCHOOL OF MATHEMATICAL SCIENCES
DEPARTMENT OF STATISTICS

M.Sc. QUANTITATIVE FINANCE
MSQF 535: Global Financial Management
LECTURE NOTES ON

Global Financial Management

PREPARED BY
NAME: N Rohit Vedhanandh
REG. NO.: 25MSQUFPY0002

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MSQF 535: GLOBAL FINANCIAL MANAGEMENT CREDITS: 4

Course Objectives:

- *The aim of this course is to provide an understanding of the fundamental concepts and managerial issues pertaining to international finance.*

Course Outcomes (COs):

CO1: Understand the global financial landscape: Comprehend the roles and functions of international financial institutions and development banks like the World Bank, IMF, and ADB, as well as the process of internationalization.

CO2: Analyze foreign exchange markets and rates: Explain the mechanisms of the foreign exchange market, including spot and forward markets, cross-rates, bid-ask spreads, and apply theories determining exchange rates.

CO3: Manage foreign exchange exposure: Evaluate and manage various types of foreign exchange exposure (translation, transaction, and economic) using different methods and strategies.

CO4: Assess multinational firm's financial management: Analyze foreign direct investment decisions, cost of capital, capital structure, budgeting, cash management, country risk, and international taxation for multinational firms.

CO5: Evaluate international financing operations: Understand the characteristics and instruments of Eurocurrency markets, interest rate and currency swaps, depository receipts (GDR and ADR), and the implications of the Euro for India.

Unit-I: International Finance Overview & Institutions

International Finance — overview — International Financial Institutions/Development Banks — World Bank — IBRD — IDA — IFC — MIGA — International Monetary Fund — Special Drawing Rights — Asian Development Bank — Internationalization process.

Unit-II: The Foreign Exchange Market

The Foreign Exchange Market — SWIFT — Arbitrage — Spot market — Forward market — Cross rates of exchange — Bid — Ask spreads — Balance of payments — Foreign exchange rates — Theories of Foreign Exchange Rate.

Unit-III: Foreign Exchange Exposure and Management

Foreign exchange exposure and management — Management of translation exposure — Methods — Management of transaction exposure — Management of economic exposure — Methods — Strategies.

Unit-IV: Financial Management of the Multinational Firm

Financial Management of the Multinational Firm — Foreign direct investment — Cost of capital and capital structure of the multinational firm — Multinational capital budgeting — Multinational cash management — Country Risk Analysis — International Taxation.

Unit-V: Financing Foreign Operations

Financing Foreign Operations — Eurocurrency markets — Instruments — Interest rate swaps — Currency swaps and its pricing — Depository receipts — GDR and ADR — Euro and its implications for India.

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COs	PO1	PO2	PO3	PO4	PO5
CO1	XXX	X	X	X	XX
CO2	XX	X	XX	X	X
CO3	X	XX	XX	X	X
CO4	XX	X	XX	XX	X
CO5	XXX	X	X	XX	XX

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Chapter 1

International Finance

1.1 International Financial Institutions

Lecture
18/08/2026

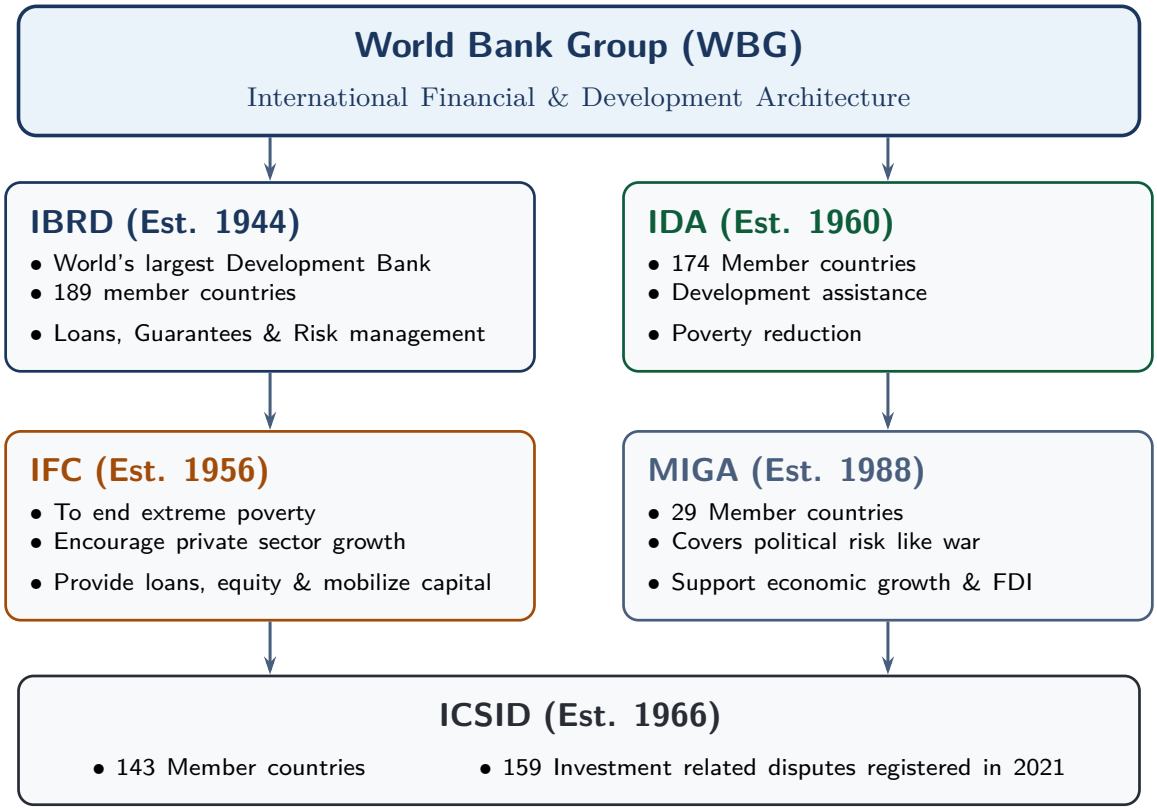


Figure 1.1: Constituent Institutions of the World Bank Group

1.1.1 IBRD (International Bank for Reconstruction and Development)

Institutional Profile: IBRD

- World's largest Development Bank.
- **Membership:** 189 countries member in this organisation.

Mission:

- Providing loans.

- Guarantee, Risk management products, advisory services for middle income and credit worthy and low income countries.
- Coordination responses to regional and global challenges.

1.1.2 IDA (International Development Association)

Institutional Profile: IDA

- **Established:** Formed in 1960.
- **Purpose:** Development assistance and poverty reduction.
- **Membership:** 174 Member of countries.

1.1.3 IFC (International Finance Corporation)

Institutional Profile: IFC

- **Established:** Established on 1956.

Mission:

- To end extreme poverty.
- To encourage private sector growth in developing countries.

Objective:

- **Investment money:** Provide loans and equity investment.
- **Mobilize capital:** Through commercial bank.
- **Provide advice:** Related to better policies, high environmental and social standards.

1.1.4 MIGA (Multilateral Investment Guarantee Agency)

Institutional Profile: MIGA

- **Established:** Established in 1988.
- **Membership:** 29 countries are member in this organisation.

Motives:

- MIGA covers political risk like war.

Objective:

- Support economic growth by encouraging FDI.
- Provides advisory services to help the developing countries.

1.1.5 ICSID (International Centre for Settlement of Investment Disputes)

Institutional Profile: ICSID

- **Established:** Established on 1966.
- **Membership:** It had 143 Member countries.
- **Disputes:** Total disputes registered on 2021, almost 159 investment related disputes.

1.2 World Bank: Operations, Funds, and Advisory Services

Lecture
21/08/2026

1.2.1 Core Operational Areas

1. Agriculture and rural development
2. Economic policy
3. Education
4. Energy
5. Environment
6. Financial sector
7. Swift system
8. Health, nutrition
9. Information, telecommunication
10. Law and justice
11. Trade related issues
12. Social protection
13. Private sector
14. Trade
15. Water resource
16. Sanitation

1.2.2 Fund-Related Operations

Fund Generation:

- IBRD
- IDA
- World Bank Treasury

Through Loans:

- Investment loans
- Development policy loans

Grants, Relief, and Targeted Initiatives:

- Provide grants
- Provide relief on debt
- For improving sanitation and water supply
- Support of vaccination or immunisation program
- HIV and AIDS pandemic
- Initiate to cut the green house effect

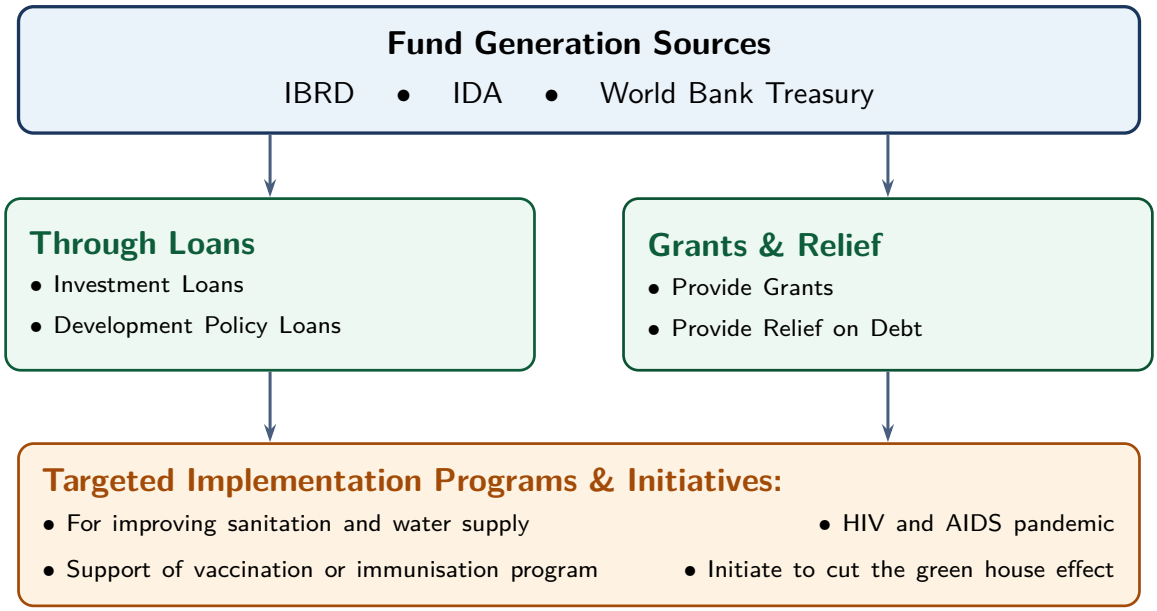


Figure 1.2: World Bank Fund Related Operations Flowchart

1.2.3 Analytics and Advisory Services

- Analytical support for poverty assessment
- Public expenditure reviews
- Country economic memoranda
- Social and structural reviews
- Sector reports
- Topics in development
- Capacity building
- Advisory service
- Global development learning network
- Global and regional programs

1.3 Governance and Bank Related Activities

1.3.1 Governance and Sovereign Representation

Lecture
24/08/2026

- Ministry of finance is the representative of world bank held once in a year.
- **Largest shareholder:** US, UK, France, Japan, Germany.
- President is elected by board of governors of 5 shareholders.

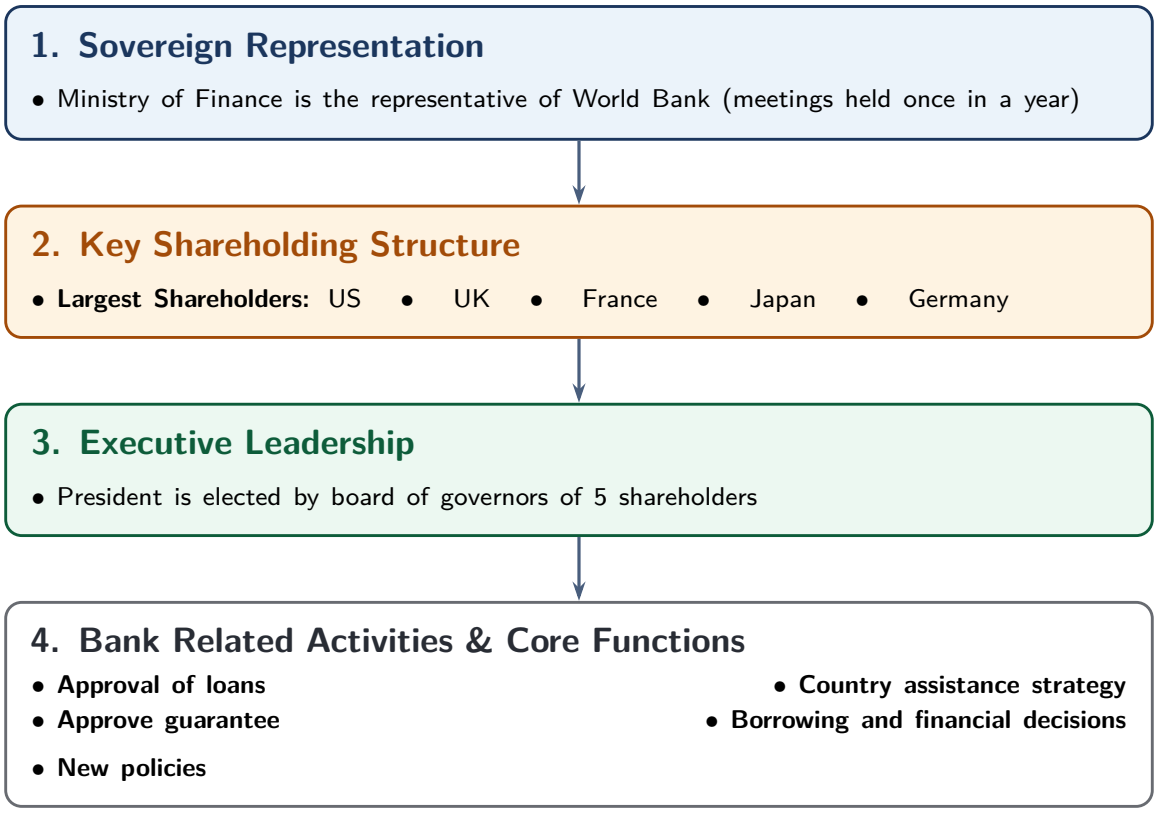


Figure 1.3: World Bank Governance Structure and Core Activities

1.3.2 Bank-Related Core Activities

- Approval of loans
- Approve guarantee
- New policies
- Country assistance strategy
- Borrowing and financial decisions

1.4 International Monetary Fund (IMF) & Sustainable Development Goals

Lecture
31/08/2026

Institutional Profile: International Monetary Fund (IMF)

The **International Monetary Fund (IMF)** was established in **1944** at the Bretton Woods Conference to foster global monetary cooperation, secure financial stability, facilitate international trade, promote high employment and sustainable economic growth, and reduce poverty around the world.

1.4.1 Core Statutory Objectives of the IMF

The International Monetary Fund operates under six foundational statutory objectives:

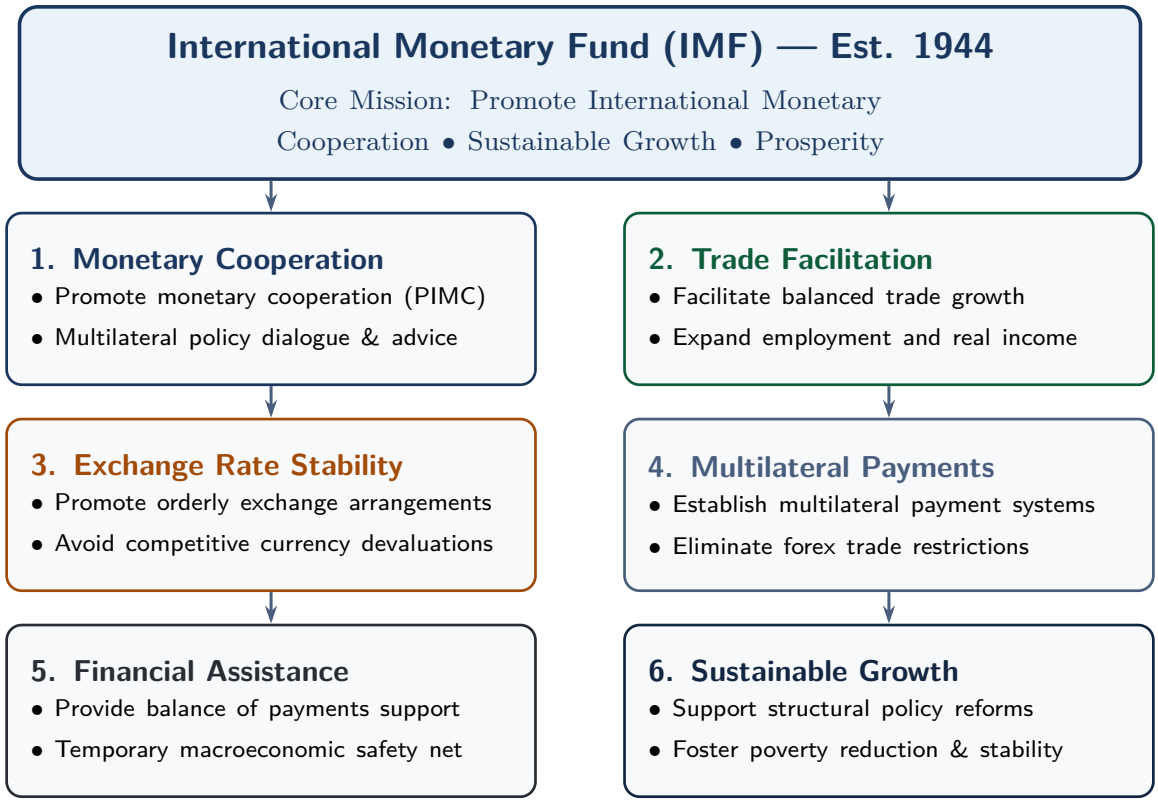


Figure 1.4: Core Strategic Objectives of the International Monetary Fund

1. **Promote International Monetary Cooperation (PIMC):** Providing the primary institutional machinery for consultation, dialogue, and collaboration on international monetary problems.
2. **Facilitate International Trade:** Promoting the expansion and balanced growth of international trade, contributing to high levels of employment and real income generation.
3. **Promote Exchange Rate Stability:** Maintaining orderly exchange arrangements among member states and avoiding competitive exchange depreciation.
4. **Establish Multilateral Payment Systems:** Assisting in establishing a multilateral system of payments for current transactions and eliminating foreign exchange restrictions that hamper world trade.
5. **Provide Financial Assistance:** Making its general resources temporarily available under adequate safeguards to members facing balance of payments disequilibria.
6. **Support Sustainable Economic Growth:** Helping countries design and implement resilient macroeconomic policies to foster prosperity and poverty reduction.

1.4.2 United Nations Sustainable Development Goals (SDGs)

The global macroeconomic architecture collaborates closely with international institutions to achieve the **17 Sustainable Development Goals (SDGs)** adopted by the United Nations:

#	Goal Name	Core Focus Area
SDG 1	**No Poverty**	End poverty in all its forms everywhere.
SDG 2	**Zero Hunger**	End hunger, achieve food security, and promote sustainable agriculture.
SDG 3	**Good Health & Well-Being**	Ensure healthy lives and promote well-being for all at all ages.
SDG 4	**Quality Education**	Ensure inclusive, equitable education and promote lifelong learning.
SDG 5	**Gender Equality**	Achieve gender equality and empower all women and girls.
SDG 6	**Clean Water & Sanitation**	Ensure availability and sustainable management of water and sanitation.
SDG 7	**Affordable & Clean Energy**	Ensure access to affordable, reliable, sustainable modern energy.
SDG 8	**Decent Work & Growth**	Promote sustained, inclusive economic growth and decent work.
SDG 9	**Industry & Infrastructure**	Build resilient infrastructure, foster innovation and industrialization.
SDG 10	**Reduced Inequalities**	Reduce income and opportunity inequality within and among countries.
SDG 11	**Sustainable Cities**	Make cities and human settlements inclusive, safe, and resilient.
SDG 12	**Responsible Consumption**	Ensure sustainable consumption and production patterns.
SDG 13	**Climate Action**	Take urgent action to combat climate change and its impacts.
SDG 14	**Life Below Water**	Conserve and sustainably use oceans, seas, and marine resources.
SDG 15	**Life on Land**	Protect, restore, and promote sustainable terrestrial ecosystems.
SDG 16	**Peace, Justice & Institutions**	Promote peaceful, inclusive societies and build accountable institutions.
SDG 17	**Partnerships for the Goals**	Strengthen the means of implementation and global partnerships.

Table 1.1: The 17 United Nations Sustainable Development Goals (SDGs)

1.5 Balance of Payments (BOP) & IMF Macroeconomic Functions

Lecture
01/09/2026

Definition 1.5.1: Balance of Payments (BOP)

The **Balance of Payments (BOP)** is a systematic statistical record of all economic and financial transactions conducted between the **residents** of a country and the **rest of the world** during a specified time period (typically a quarter or a financial year).

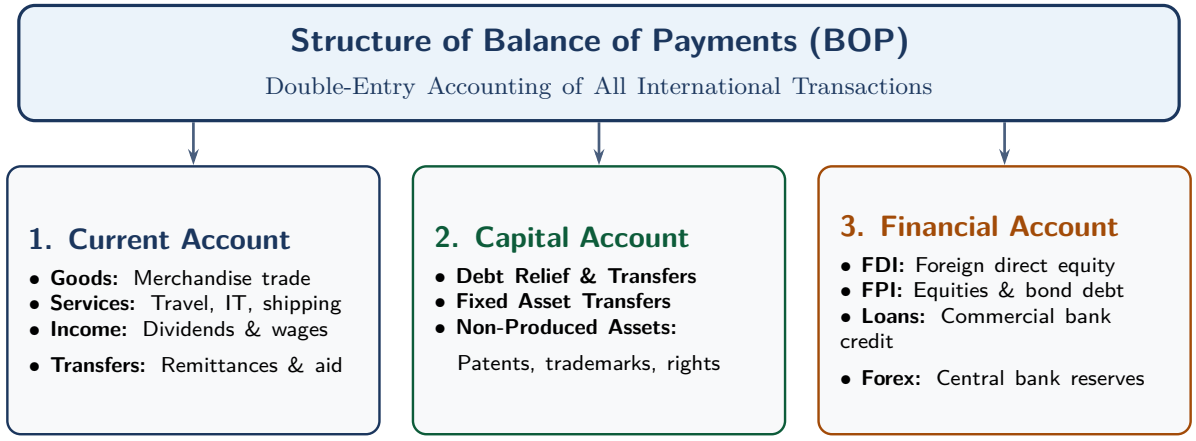


Figure 1.5: Constituent Accounts of the Balance of Payments Framework

1.5.1 Structure and Components of BOP Accounts

The Balance of Payments is structured into three primary interrelated accounts:

1. **Current Account:** Records trade in goods and services, investment income, and unilateral transfers:
   - **Export and Import of Goods:** Visible / merchandise trade balance.
   - **Export and Import of Services:** Invisible trade (software services, banking, shipping, tourism).
   - **Primary Income (Investment Returns):** Cross-border investment returns (dividends, interest receipts/payments) and employee wages.
   - **Current Transfers (Secondary Income):** Unrequited unilateral flows including personal remittances, grants, donations, and gifts.
2. **Capital Account:** Records net international capital transfers and non-financial asset movements:
   - **Capital Transfers:** Debt forgiveness, sovereign grants, and migrant asset transfers.
   - **Transfer of Fixed Asset Ownership:** Cross-border transfers of sovereign or institutional fixed assets.
   - **Acquisition / Disposal of Non-Produced, Non-Financial Assets:** Rights to natural resources, patents, trademarks, brand leases, and copyrights.
3. **Financial Account:** Records international transactions in financial assets and liabilities:
   - **Foreign Direct Investment (FDI):** Long-term controlling equity investments in foreign enterprises ($> 10\%$ voting power).

- **Foreign Portfolio Investment (FPI):** Cross-border investments in equity shares, corporate debt, and sovereign bonds for yield rather than managerial control.
- **Other Investments:** Commercial bank lending, official bilateral loans, and trade credits.
- **Changes in Foreign Exchange Reserves:** Official reserve asset transactions managed by the Central Bank to stabilize the currency.

1.5.2 BOP Disequilibrium: Surplus vs. Deficit

- **BOP Surplus:** Occurs when autonomous international receipts exceed total autonomous payments:

$$\text{Total Receipts} > \text{Total Payments} \implies \text{Net Inflow of Foreign Currency / Reserve Accumulation} \tag{1.1}$$

- **BOP Deficit:** Occurs when autonomous international payments exceed total receipts:

$$\text{Total Payments} > \text{Total Receipts} \implies \text{Net Outflow of Reserves / External Debt Expansion} \tag{1.2}$$

1.5.3 Causes and Remedial Measures for BOP Deficits

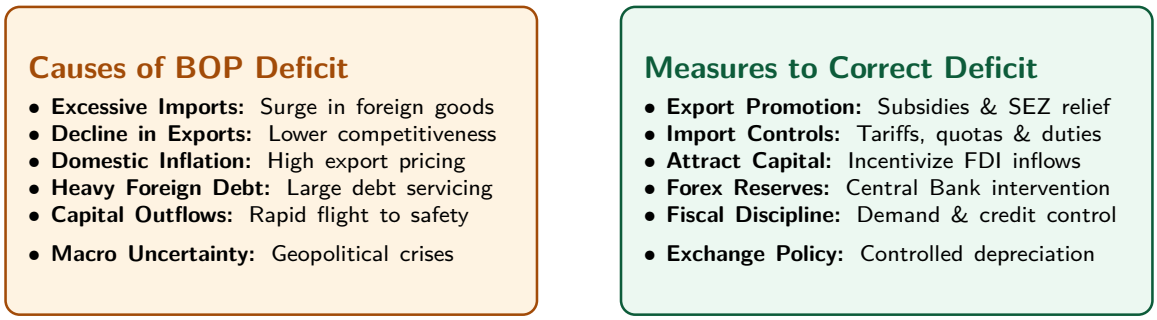


Figure 1.6: Macroeconomic Dynamics: Causes and Policy Correctives for BOP Deficit

1.5.4 Core Macroeconomic Functions of the IMF

The International Monetary Fund executes its mandate through three fundamental pillars of macroeconomic intervention:

1. **Surveillance (Economic Health Monitoring):**
   - Conducts bilateral Article IV consultations with member nations.
   - Monitors global, regional, and national economic policies, exchange rate regimes, and systemic financial vulnerabilities.
2. **Financial Assistance (Lending):**
   - Provides balance of payments crisis loans and precautionary credit lines to countries facing external liquidity shocks.
   - Conditions loans on economic stabilization programs and structural policy reforms.
3. **Capacity Development (Technical Assistance & Training):**

- Delivers specialized capacity building and advisory services to central banks, ministries of finance, and statistical agencies.
- Modernizes tax administration, monetary policy frameworks, banking supervision, and macroeconomic data systems.

1.6 Special Drawing Rights (SDR)

Lecture
02/09/2026

Institutional Profile: Special Drawing Rights (SDR)

The **Special Drawing Right (SDR)** is an international supplementary foreign exchange reserve asset created by the **International Monetary Fund (IMF)** in **1969**.

- **Nature:** The SDR is **not a currency** and not a claim on the IMF itself.
- **Function:** It represents a potential claim on the freely usable currencies of IMF member states.

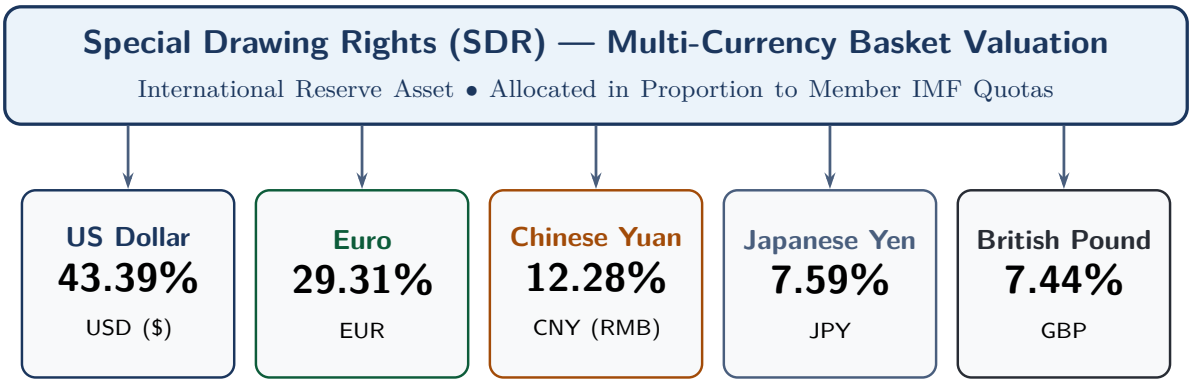


Figure 1.7: Composition and Weighting of the IMF Special Drawing Rights (SDR) Currency Basket

1.6.1 Purpose and Functions of SDR

- **Supplement Official Foreign Exchange Reserves:** Augmenting member countries' gold and foreign exchange reserve assets.
- **Provide International Liquidity:** Supplying unconditional liquidity to the global financial system during liquidity shortages.
- **Strengthen External Financial Position:** Enhancing sovereign balance sheet resilience without increasing debt distress.
- **Help Countries Meet BOP Needs:** Assisting nations during severe external balance of payments distress.

1.6.2 SDR Valuation Basket and Weightage

The value of the SDR is determined daily by the IMF on the basis of a trade- and finance-weighted basket of five major global currencies. The relative weights are reviewed every five years:

Currency	ISO Code	Central Bank / Authority	Approx. Weightage
US Dollar	USD	Board of Governors of the Federal Reserve System	**43.39%**
Euro	EUR	European Central Bank (Eurosystem)	**29.31%**
Chinese Yuan (Renminbi)	CNY	People's Bank of China	**12.28%**
Japanese Yen	JPY	Bank of Japan	**7.59%**
British Pound Sterling	GBP	Bank of England	**7.44%**

Table 1.2: Constituent Currencies and Weightages of the SDR Basket

1.6.3 Allocation and Interest Mechanism of SDRs

- **Allocation of SDRs:** SDRs are allocated to IMF member nations in direct proportion to their existing **IMF Quota shares**. Members can hold SDRs as part of their official reserves or exchange them with other members for freely usable currencies.
- **SDR Interest Rate (SDRi):** The SDR interest rate is calculated weekly based on a weighted average of representative short-term interest rates in the money markets of the SDR basket currencies.

1.6.4 Core Strategic Objectives of SDRs

1. **Increase Foreign Exchange Reserves:** Bolsters national central bank reserve adequacy metrics.
2. **Improve Liquidity During Crisis:** Infuses non-debt-creating global liquidity during systemic macroeconomic shocks.
3. **Reduce Pressure on Forex Reserves:** Decreases reliance on expensive domestic borrowing or volatile external debt markets.
4. **Support Balance of Payments (BOP):** Bridges temporary external account deficits.

1.7 Asian Development Bank (ADB) & Institutional Comparison

Lecture
02/09/2026

Institutional Profile: Asian Development Bank (ADB)

- **Established:** 1966.
- **Headquarters:** Manila, Philippines.
- **Nature of Institution:** Multilateral Development Bank (MDB).
- **Geographic Scope:** Asia and the Pacific Region.
- **India's Membership:** Founding member of ADB since 1966.

1.7.1 Core Objectives and Operational Mandate of ADB

- **Reduce Poverty:** Eradicating extreme poverty across member developing nations.
- **Promote Economic Growth:** Fostering sustainable, inclusive economic progress.
- **Support Infrastructure Development:** Financing critical capital assets (transportation, energy, telecommunications).

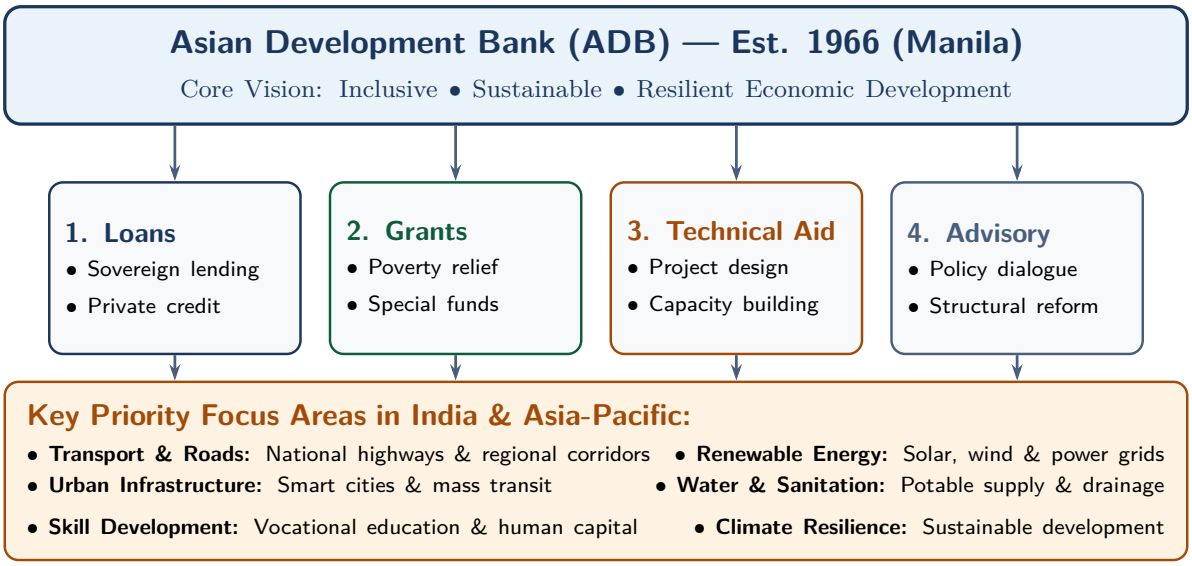


Figure 1.8: Asian Development Bank: Operational Channels and Priority Development Areas

- **Encourage Regional Cooperation:** Deepening cross-border economic ties and regional trade integration.
- **Promote Environmental Sustainability:** Addressing climate change mitigation and adaptation.
- **Improve Social Development:** Investing in education, healthcare, and gender equity.

1.7.2 ADB and India Partnership Focus Areas

India is a founding member of the ADB (since 1966) and one of its largest borrowers. The partnership focuses on six key pillars:

1. **Urban Infrastructure:** Water management, sewerage, and smart urban mobility.
2. **Transport and Roads:** State highways, rural connectivity, and economic corridors.
3. **Renewable Energy:** Solar parks, green energy corridors, and transmission modernization.
4. **Water and Sanitation:** Irrigation modernization and clean drinking water access.
5. **Skill Development:** Industrial training institutes and youth workforce readiness.
6. **Climate Resilient Development:** Coastal protection and climate adaptation programs.

1.7.3 Comparative Analysis: ADB vs. IMF

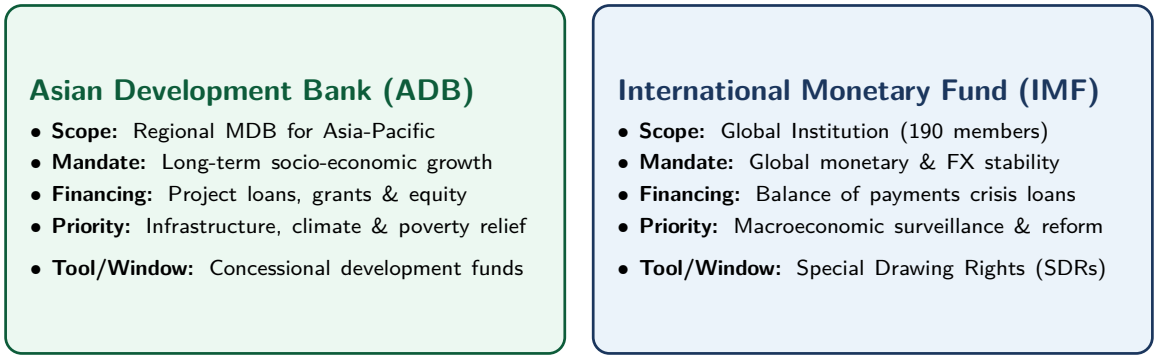


Figure 1.9: Institutional Comparison: Asian Development Bank (ADB) vs. International Monetary Fund (IMF)

Dimension	Asian Development Bank (ADB)	International Monetary Fund (IMF)
Institutional Nature	Regional Multilateral Development Bank	Global Monetary Authority
Geographic Scope	Asia and the Pacific Region	Global (Worldwide membership)
Primary Objective	Long-term economic development and poverty reduction	Global monetary stability and BOP crisis resolution
Financing Structure	Project-specific capital loans, equity, and grants	Short-to-medium-term balance of payments support
Core Instruments	Sovereign/non-sovereign project loans, technical aid	Stand-by Arrangements, EFF, SDRs

Table 1.3: Summary Matrix: Key Differences between ADB and IMF

1.8 Internationalisation Process

Lecture
03/09/2026

Definition 1.8.1: Internationalisation

Internationalisation is the strategic process of expanding a company's business operations, products, and services into foreign markets across international borders.

1.8.1 Stages of Internationalisation

The evolution of a domestic firm into a globally integrated enterprise typically progresses through five distinct developmental stages:

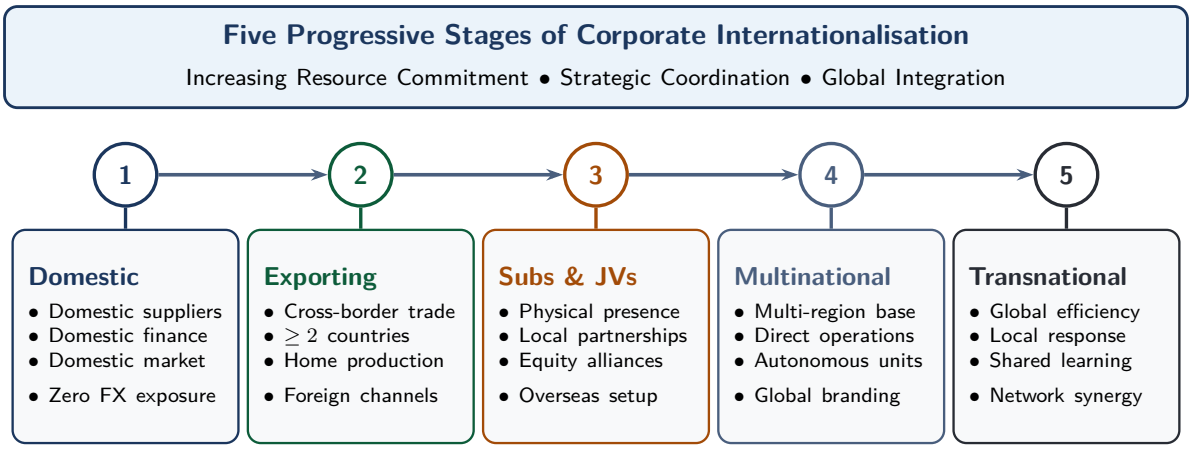


Figure 1.10: The Five Progressive Stages of the Corporate Internationalisation Process

1. **Domestic Operations:** Domestic companies focus exclusively on domestic suppliers, domestic financial institutions, and domestic market opportunities without active foreign exposure.
2. **Export Operations:** In this stage, the firm expands sales beyond the home country; while engaging in export activities, business operations span across two or more countries.
3. **Subsidiaries and Joint Ventures:** The business establishes physical partnerships, localized manufacturing units, or overseas operational offices to secure local market footprint.
4. **Multinational Operations:** Operations are built and coordinated across multiple geographic regions, operating with substantial international assets and multi-country facilities.
5. **Transnational Operations:** The firm operates as an integrated global network, dynamically balancing global scale efficiency with local responsiveness across all national markets.

1.8.2 Motivations for International Expansion

Firms undertake cross-border expansion driven by key strategic imperatives:

- **Market Growth (Domestic Saturation):** When domestic markets become saturated, expanding overseas unlocks new customer bases and sustained sales volume.
- **Cost Reduction:** Achieving economies of scale and operational efficiencies by spreading overhead and fixed costs over larger global production volumes.
- **Competitive Pressures:** Responding to global competitors entering domestic turf or defending worldwide market share by establishing presence in rival home markets.
- **Customer Following (Suppliers Follow Their Customers):** Specialized suppliers expand internationally to maintain seamless service relationships with their multinational corporate buyers.
- **Attractive Cost Structures Globally:** Leveraging cross-border cost disparities, such as lower labor costs, cheaper raw materials, and regional tax and infrastructure incentives.
- **Growth Rate and Potential:** Capturing growth opportunities in high-potential emerging economies experiencing rapid industrial and consumption expansion.

1.8.3 Foreign Market Entry Strategies

Entering international markets involves selecting an appropriate entry mode that balances operational control, resource commitment, and financial risk:

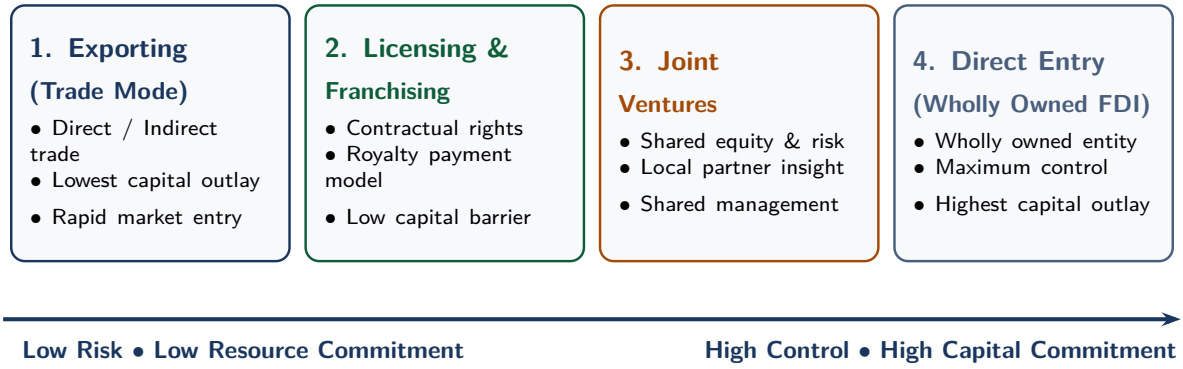


Figure 1.11: Strategic Spectrum of Foreign Market Entry Modes

1. **Exporting:** Selling domestically produced goods directly or indirectly through foreign intermediaries, minimizing initial capital exposure.
2. **Licensing and Franchising:** Granting foreign partners legal rights to intellectual property, trademarks, or operational systems in return for fees and royalties.
3. **Joint Ventures:** Forming an independent commercial entity jointly owned with a domestic or international partner to share capital, risks, and local market expertise.
4. **Direct Investment (FDI):** Committing direct equity capital to establish wholly-owned foreign subsidiaries or acquire local operating enterprises, securing maximum operational control.

1.8.4 Strategic Advantages and Disadvantages of Internationalisation

Expanding globally yields significant strategic benefits while exposing multinational enterprises to multidimensional environmental risks:

Strategic Advantages	Operational & Market Disadvantages
High Living Standards: Access to diversified, high-quality consumer products and advanced services bolsters global living standards.	**Political Factors:** Vulnerability to foreign government policies, geopolitical friction, expropriation risk, and regulatory changes.
Wider Market: Broadens addressable customer reach beyond saturated domestic boundaries, accelerating sales growth.	**Exchange Instability:** Unhedged currency volatility generates translation, transaction, and economic exposure.
Large Scale Economies: Global manufacturing, procurement, and distribution generate substantial economies of scale.	**Tariffs, Quotas & Trade Barriers:** Protectionist cross-border tariffs, quantitative import quotas, and non-tariff friction.
Cultural Transition: Enhances corporate cross-cultural knowledge, managerial flexibility, and global workforce synergy.	**Technological Challenges:** Cross-border technology transfer costs, disparities in technical infrastructure, and intellectual property theft.
Optimum Utilisation: Ensures optimal worldwide allocation of capital, specialized labor, and productive physical resources.	**Managerial Complexity:** Escalating communication overhead, coordination latency, and multi-jurisdictional compliance costs.

Table 1.4: Summary Matrix: Advantages and Disadvantages of Corporate Internationalisation

1.9 Emerging Challenges in International Finance

Lecture
07/09/2026

International financial operations and multinational corporate management operate within a dynamic global environment characterized by diverse macroeconomic, regulatory, geopolitical, and operational complexities. The twenty primary emerging challenges identified in modern international finance encompass:

1. **Economic System Disparities:** Navigating structural differences across capitalist, socialist, and mixed economic systems with differing degrees of state intervention, price controls, and private property rights.
2. **Tariff and Non-Tariff Barriers:** Managing import duties, customs tariffs, quantitative quotas, import licensing restrictions, and stringent technical standards imposed by sovereign trading partners.
3. **Political Risk:** Protecting foreign assets and cash flows against sovereign policy reversals, discriminatory regulations, expropriation, nationalisation, and civil unrest.
4. **Environmental Safeguards:** Adhering to diverse cross-border environmental protection norms, carbon emissions caps, green compliance standards, and sustainable operational requirements.
5. **Dumping Practices:** Mitigating predatory export pricing where foreign competitors export merchandise at prices below production cost or domestic market values, triggering anti-dumping duties.
6. **Cultural Differences:** Managing distinct consumer habits, workplace values, negotiating styles, and organizational cultures across host countries.

7. **Language Differences:** Overcoming cross-border linguistic barriers in legal contract formulation, executive communication, marketing campaigns, and commercial negotiations.

8. **Intellectual Property Rights (IPR):** Safeguarding proprietary patents, trademarks, industrial designs, and trade secrets against copyright infringement and reverse engineering in foreign jurisdictions.

9. **Cyber Crimes:** Defending international digital payment gateways, cloud banking networks, and proprietary transactional data against ransomware, cyber extortion, and cross-border financial fraud.

10. **Transfer Pricing:** Managing internal transaction pricing for goods, services, and intangibles transferred between related corporate entities across differing national tax jurisdictions.

11. **International Taxation:** Handling complex double taxation avoidance agreements (DTAA), withholding taxes on cross-border dividends and royalties, and Base Erosion and Profit Shifting (BEPS) regulations.

12. **Economic Crisis and Currency Crisis:** Insulating corporate balance sheets against systemic macroeconomic downturns, sudden capital flight, sovereign debt defaults, and precipitous foreign exchange collapses.

13. **Interest Rate Volatility:** Managing exposure to divergent monetary policy cycles, benchmark interest rate fluctuations, and cross-border interest rate differentials.

14. **Foreign Exchange Risk:** Mitigating transaction exposure (contracted receivables/payables), translation exposure (accounting consolidation of foreign subsidiaries), and economic exposure (long-term cash flow competitiveness).

15. **Cold War Tensions and Geopolitical Friction:** Navigating bilateral diplomatic standoffs, trade embargoes, economic sanctions, supply chain weaponisation, and regional conflicts between nations.

16. **International Business Cycles:** Addressing asynchronous economic growth rates, localized recessions, and cross-border cyclical propagation across global markets.

17. **Operational Cash Management:** Optimizing global liquidity, managing multi-currency treasury balances, repatriating blocked funds, and executing centralized cash pooling.

18. **International Terrorism:** Safeguarding overseas physical infrastructure, logistical corridors, and personnel against terrorist threats and regional destabilization.

19. **Creditworthiness Assessment:** Evaluating counterparty default risk, sovereign risk ratings, and foreign financial institution solvency in international credit transactions.

20. **Method of Payment Selection:** Structuring cross-border commercial settlements (advance payment, documentary letters of credit, collections, open account) to reconcile importer-exporter liquidity and default risks.

Chapter 2

The Foreign Exchange Market

2.1 Structure and Participants of the Foreign Exchange Market

Lecture
07/09/2026

Definition 2.1.1: Foreign Exchange Market

The **Foreign Exchange (Forex) Market** is the global organizational and institutional framework where the currency of one country is exchanged for the currency of another country.

In the foreign exchange market, currencies are quoted against a single base unit of another currency (e.g., 1 USD = 95.88 INR). The market operates on a continuous, decentralized over-the-counter (OTC) basis across global financial centers.

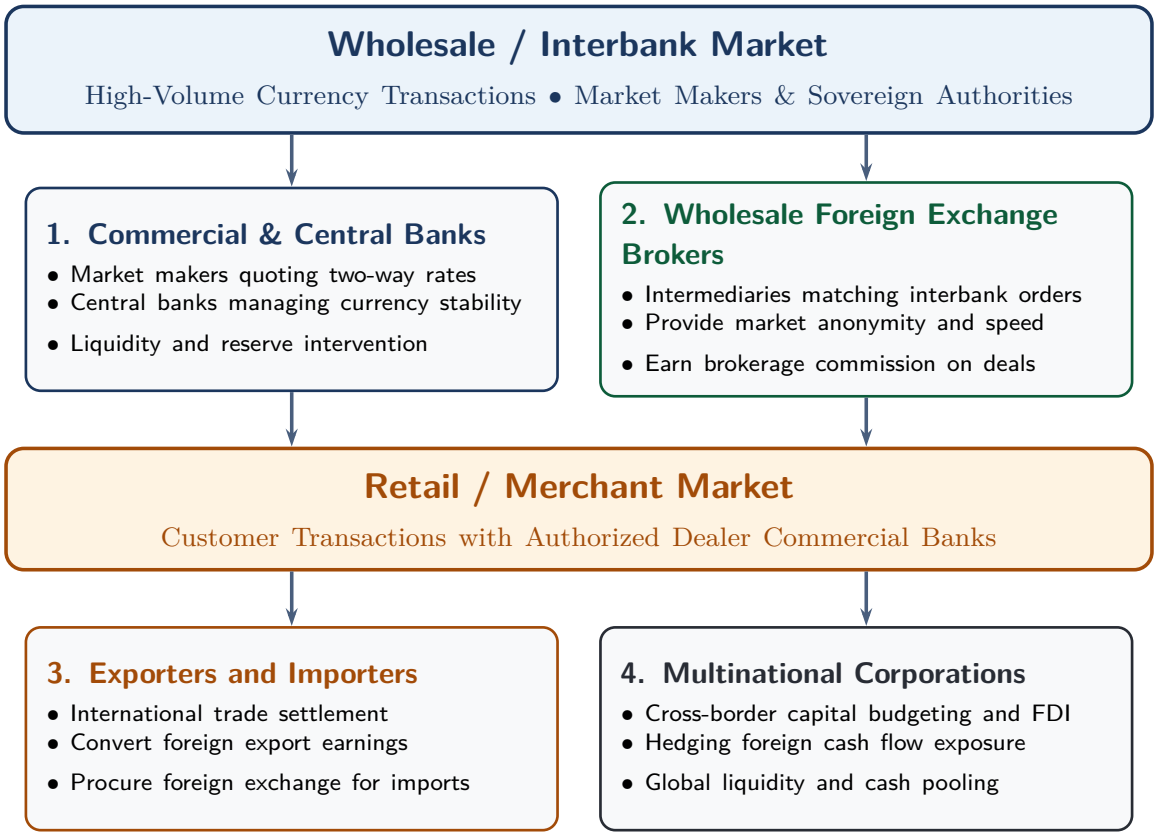


Figure 2.1: Architecture and Core Participants of the Foreign Exchange Market

2.1.1 Core Market Participants

The foreign exchange market operates through four constituent participant categories:

1. **Commercial Banks and Central Banks:** Authorized dealer commercial banks serve as market makers by quoting two-way prices (bid and ask). Central banks participate to execute sovereign monetary policy, hold national reserves, and stabilize volatile domestic exchange rates.
2. **Wholesale Foreign Exchange Brokers:** Intermediaries operating between commercial banks in the interbank tier, matching buy and sell orders rapidly while preserving market anonymity.
3. **Exporters and Importers:** Commercial entities requiring foreign exchange conversion to settle merchandise and service trade transactions across international borders.
4. **Multinational Corporations (Corporates):** Global enterprises transferring capital across jurisdictions for direct investment, cross-border dividend repatriation, and liquidity hedging.

2.2 Foreign Exchange Quotation Mechanics and Settlement Horizons

Lecture
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Definition 2.2.1: Foreign Exchange Quotation

A **Foreign Exchange Quotation** is the price of one currency expressed in units of another currency.

2.2.1 Delivery Horizons: Spot vs. Forward Rates

Foreign exchange rates are quoted depending on the scheduled delivery date of the underlying currencies:

- **Spot Rate:** The exchange rate quoted for immediate delivery and settlement of currencies. In international banking practice, spot delivery is settled **two working days** after the date of the deal ($T + 2$).
- **Forward Rate:** The exchange rate agreed upon today for the exchange of currencies at a specified future date (exceeding two working days from the transaction date).

2.2.2 Types of Exchange Rate Quotations

1. **Direct Quotation (Home Currency Quotation):** Expresses the price of **one unit of foreign currency** in terms of variable units of home currency.

$$\text{Home Currency (Variable)} = 1 \text{ Unit of Foreign Currency (Fixed)} \tag{2.1}$$

   Example in India: $95.88\,\text{INR} = 1\,\text{USD}$ represents a direct quotation for the US Dollar in India.

2. **Indirect Quotation (Foreign Currency Quotation):** Keeps the home currency fixed at **one unit** and expresses the exchange rate as variable units of foreign currency.

$$1 \text{ Unit of Home Currency (Fixed)} = \text{Foreign Currency (Variable)} \tag{2.2}$$

   Example: $1\,\text{USD} = 0.87\,\text{EUR}$ represents an indirect quotation from the US domestic perspective.

2.2.3 Bid-Ask Spread and Percentage Spread

Foreign exchange quotes consist of a two-way price quoted by market makers:

- **Bid Price:** The price at which the bank/market maker is willing to **buy** foreign currency.
- **Ask Price (Offer):** The price at which the bank/market maker is willing to **sell** foreign currency.
- **Spread:** The gross margin earned by the quoting dealer:

$$\text{Spread} = \text{Ask Price} - \text{Bid Price} \tag{2.3}$$

- **Percentage Spread:** The proportional transaction cost relative to the asking price:

$$\text{Percentage Spread } (\%) = \frac{\text{Ask Price} - \text{Bid Price}}{\text{Ask Price}} \times 100 \tag{2.4}$$

Example 2.2.1: Bid-Ask Spread Calculation

Problem Statement:
In the interbank foreign exchange market, the US Dollar is quoted as:

$$1\,\text{USD} = \text{Rs. } 95.0000 - 95.0050$$

Calculate the absolute spread and the percentage spread for this quotation.

Solution:
From the quotation:

$$\text{Bid Price} = \text{Rs. } 95.0000$$
$$\text{Ask Price} = \text{Rs. } 95.0050$$

1. Absolute Spread:

$$\text{Spread} = \text{Ask} - \text{Bid} = 95.0050 - 95.0000 = \text{Rs. } 0.0050$$

2. Percentage Spread:

$$\begin{aligned}\text{Percentage Spread} &= \frac{\text{Ask} - \text{Bid}}{\text{Ask}} \times 100 \\ &= \frac{0.0050}{95.0050} \times 100 = 0.00526\%\end{aligned}$$

Note: If the quotation were 95.0000 − 95.0140, the absolute spread would be 0.0140, yielding a percentage spread of approximately 0.014%.

2.2.4 Cross Rates

Definition 2.2.2: Cross Rate

A **Cross Rate** is the exchange rate between two non-home currencies, derived through their respective exchange rates against a common vehicle currency (typically the US Dollar).

Example: In the Indian market where INR is the domestic currency, quotes for EUR/USD, GBP/USD, or JPY/CNY represent cross-rate transactions.

2.2.5 Settlement Horizons

Depending on the time elapsed between the transaction date (T) and the physical settlement date, deals are categorized into four standard settlement horizons:

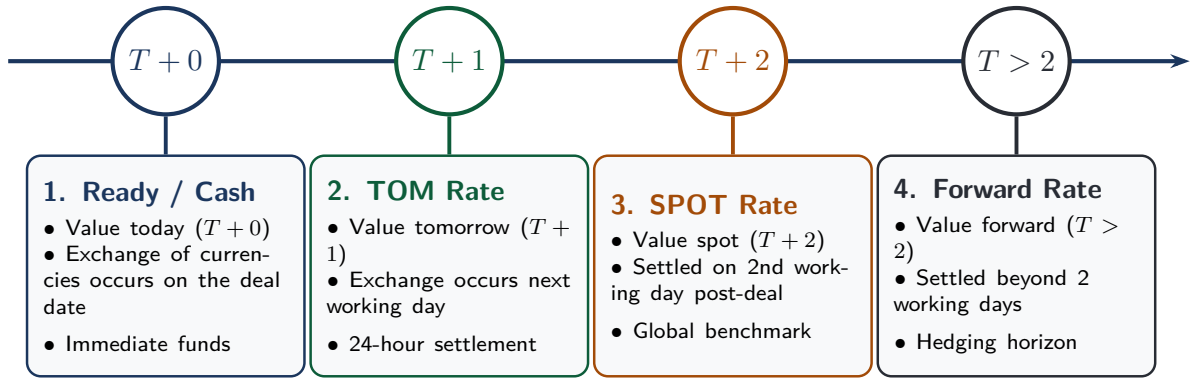


Figure 2.2: Chronological Spectrum of Foreign Exchange Settlement Horizons

1. **Cash / Ready Rate ($T+0$):** Settlement and physical delivery of the currencies occur on the very date of the deal.
2. **TOM Rate ($T+1$):** Settlement takes place on the next immediate working day following the deal date.
3. **SPOT Rate ($T+2$):** Standard interbank convention where the exchange of currencies occurs on the second working day after the deal date.
4. **Forward Rate ($T>2$):** Delivery of currencies occurs at an agreed future date beyond two working days.

2.3 International Payment and Financial Telecommunication Systems

Lecture
16/09/2026

Cross-border interbank settlements and foreign currency transfers rely on specialized international telecommunications and multilateral clearing systems.

SWIFT (Est. Brussels)

- **Nature:** Bank-owned financial messaging cooperative
- **Ownership:** Owned by >250 member banks across Europe and North America
- **Network:** Connects $>25,000$ financial institutions across 200+ nations
- **Function:** Authenticated, cryptographic transmission of financial messages
- **Identifier:** Standardized Bank Identifier Codes (BIC)
- **India Hub:** Primary regional processing centre situated in Mumbai

CHIPS (Est. 1971, New York)

- **Nature:** Multilateral electronic clearing system
- **Founders:** 12 NY Clearing House commercial banks
- **Scale:** World's largest private USD clearing system
- **Function:** Daily electronic netting and settlement
- **Paperless:** 100% electronic (zero physical checks)
- **Coverage:** Foreign exchange and USD cross-border flows

Figure 2.3: Institutional Comparison: SWIFT (Messaging) vs. CHIPS (Clearing & Settlement)

2.3.1 SWIFT (Society for Worldwide Interbank Financial Telecommunication)

- **Organizational Nature:** A bank-owned cooperative society registered in **Brussels, Belgium**.
- **Ownership & Scope:** Formed by 250 member banks in Europe and North America; links more than **25,000 financial institutions** worldwide.
- **Bank Identifier Codes (BIC):** Allots standardized identification codes (SWIFT/BIC codes) to each financial institution for routing.
- **Services:**
  - International payment instructions.
  - Account statements from SWIFT.
  - Specialized administrative and trade finance messages.
- **Indian Operations:** The primary regional processing center for India is situated in **Mumbai**.
- **Key Operational Advantages:**
  1. Provides a reliable, secure method of transmitting messages across vast global locations.
  2. Guarantees high accuracy and cryptographic authentication.
  3. Enables counterparties to respond and process payments immediately.

2.3.2 CHIPS (Clearing House Interbank Payments System)

- **Organizational Nature:** A computerized electronic payment system owned by commercial banks constituting the **New York Clearing House Association**.
- **Historical Commencement:** Began operations in **1971**.
- **Operational Scope:** Represents the largest private-sector payment clearing system in the world. It clears foreign exchange and cross-border US Dollar transactions.
- **Settlement Mechanism:** Daily electronic netting and settlement of payments and receipts without physical checks (purely paperless interbank settlement).

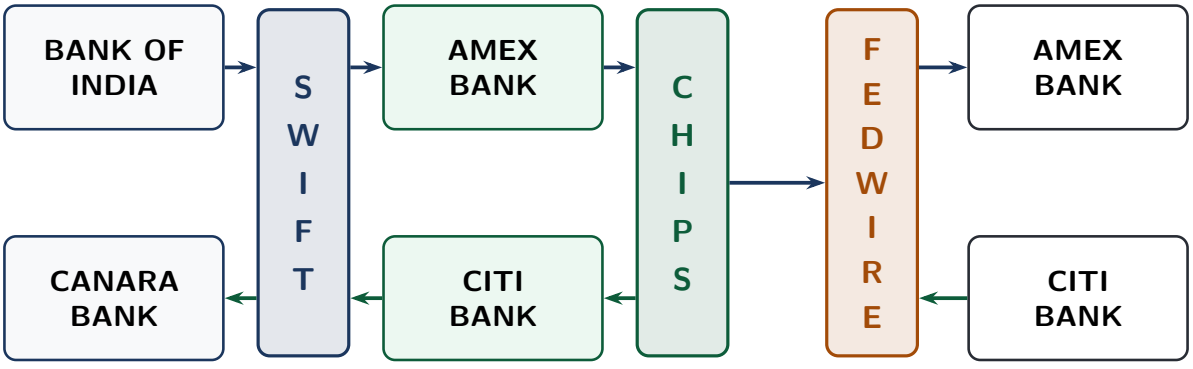


Figure 2.4: Operational Mechanism for Cross-Border Settlement through CHIPS, SWIFT, and Fedwire

2.4 Market Terminologies, Margins, and Transaction Classifications

2.4.1 Key Market Terminologies

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- **Market Maker:** Major commercial banks that stand ready to buy and sell currencies at publicly quoted prices, providing two-way market liquidity.
- **Two-Way Quotation:** A quotation indicating two distinct prices:
  - Price at which the market maker is willing to **buy** foreign currency (Bid).
  - Price at which the market maker is willing to **sell** foreign currency (Ask).
- **Forward Margin / SWAP Points:** The difference between the forward exchange rate and the spot exchange rate.
- **Forward Premium:** Occurs when foreign currency is costlier under the forward rate than under the spot rate (Forward Rate > Spot Rate).
- **Forward Discount:** Occurs when foreign currency is cheaper for forward delivery than for spot delivery (Forward Rate < Spot Rate).
- **Base Rate:** The ongoing interbank exchange rate on the basis of which commercial banks calculate merchant customer rates.

2.4.2 Golden Rules: The Banker's Perspective

When analyzing any foreign exchange transaction, two fundamental principles must be strictly observed:

The Two Foundational Principles of Foreign Exchange Accounting

1. **Banker's Point of View:** All transactions are designated strictly from the perspective of the **bank**, not the customer.
2. **Underlying Asset:** The item being bought or sold is always the **Foreign Currency**.

Consequently, transactions are classified as:

- **Purchase Transaction:** The bank has **purchased foreign currency** (the bank receives foreign currency and pays home currency/INR to the customer).
- **Sale Transaction:** The bank has **sold foreign currency** (the bank parts with foreign currency and receives home currency/INR from the customer).

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Example 2.4.1: Classification of Foreign Exchange Transactions

Problem Statement:
Based on the banker's perspective, determine whether each of the following transactions constitutes a **Purchase** or a **Sale** of foreign exchange:

1. The bank issues a Demand Draft (DD) on London for 100 EUR.
2. A customer of the bank purchases a Telegraphic Transfer (TT) on New York for 500 EUR.

3. A traveller encashes a traveller's cheque for 50 EUR at the bank counter.
4. The bank purchases a Demand Draft (DD) drawn on London for 500 EUR.

Solution:

1. **Sale of Foreign Exchange:** In issuing a DD on London, the bank sells foreign currency (100 EUR) to the customer and receives rupees.
2. **Sale of Foreign Exchange:** By executing a TT remittance to New York, the bank parts with foreign currency (500 EUR) and receives rupees from the remitter.
3. **Purchase of Foreign Exchange:** When encashing the traveller's cheque, the bank acquires/buys foreign currency (50 EUR) and pays equivalent rupees to the traveller.
4. **Purchase of Foreign Exchange:** By purchasing a draft drawn on London, the bank buys foreign currency (500 EUR) from the presenter.

2.4.3 Calculation of Forward Rates under Direct Quotation

In the interbank market, forward margins are quoted as points:

- **Ascending Order Margins $\implies$ Premium:** If forward margins are quoted in ascending order (e.g., 0.2000/0.2100), foreign currency is at a **premium**. Margins are **added** to the spot rate:

$$\text{Forward Rate} = \text{Spot Rate} + \text{Forward Premium} \tag{2.5}$$

- **Descending Order Margins $\implies$ Discount:** If forward margins are quoted in descending order (e.g., 0.6000/0.5700), foreign currency is at a **discount**. Margins are **deducted** from the spot rate:

$$\text{Forward Rate} = \text{Spot Rate} - \text{Forward Discount} \tag{2.6}$$

Example 2.4.2: Determining Forward Margins via Covered Interest Parity

Problem Statement:
The spot rate for US Dollars in the Mumbai interbank market is 1 USD = 95 INR. The annual rate of interest in India is 8% p.a., while in New York it is 5% p.a. A bank is asked to quote a three-month selling rate to a customer for 10,000 USD, assuming the entire interest differential gain or loss is passed on to the customer. Calculate the theoretical forward rate and forward premium.

Solution:
Given:

$$S = 95\,\text{INR/USD}, \quad r_{\text{INR}} = 8\% = 0.08, \quad r_{\text{USD}} = 5\% = 0.05, \quad t = \frac{3}{12} = 0.25\,\text{years}$$

1. Accounting Cash Flow Schedule (for USD 10,000):

Particulars	Amount
Purchase Dollar (Foreign Inflow)	10,000.0000 USD
Interest earned for 3 months @ 5% p.a. $(10,000 \times 0.05 \times \frac{3}{12})$	125.0000 USD
Total Foreign Currency Realization at Maturity	**10,125.0000 USD**
Borrow in Mumbai to pay for USD @ INR 95 $(95 \times 10,000)$	950,000.0000 INR
Interest payable for 3 months @ 8% p.a. $(950,000 \times 0.08 \times \frac{3}{12})$	19,000.0000 INR
Total Rupee Outflow to be Recovered (Rate Quoted)	**969,000.0000 INR**

2. Theoretical Forward Rate (Exact Covered Interest Parity):

$$F = \frac{\text{Total INR Outflow}}{\text{Total USD Realization}} = \frac{969,000.0000}{10,125.0000} = \mathbf{95.7037\,INR/USD}$$

This matches the exact formula:

$$F = S \times \left[\frac{1 + r_{\text{INR}} \times t}{1 + r_{\text{USD}} \times t}\right] = 95 \times \frac{1.0200}{1.0125} = 95.7037\,\text{INR}$$

3. Linear Interest Differential Approximation:

$$F_{\text{linear}} \approx S \times [1 + (r_{\text{INR}} - r_{\text{USD}}) \times t] = 95 \times [1 + (0.08 - 0.05) \times 0.25] = \mathbf{95.7125\,INR}$$

4. Forward Premium Difference:

$$\Delta = 95.7125 - 95.7037 = \mathbf{0.0088\,INR}$$

The forward quotation reflects a forward premium of 0.0088 INR per USD over the exact parity rate.

2.5 Indian Merchant Rates: Buying Rates and Mechanics

Lecture
18/09/2026

Merchant rates are the retail foreign exchange rates quoted by authorized dealer commercial banks to their corporate and individual customers, calculated from ongoing interbank rates as prescribed by FEDAI (Foreign Exchange Dealers' Association of India).

2.5.1 Classification of Buying Rates

Authorized dealers in India quote two standard types of buying rates:

1. **Telegraphic Transfer (TT) Buying Rate:** Applied when the transaction does not involve any delay in the bank obtaining reimbursement in foreign currency.
2. **Bill Buying Rate:** Applied when the bank purchases or discounts clean or documentary commercial trade bills.

2.5.2 TT Buying Rate Mechanics

The TT Buying Rate is applied in three distinct operational situations:

1. Payment of foreign Demand Drafts (DD), Mail Transfers (MT), and Telegraphic Transfers (TT) drawn on the quoting bank.
2. Payment of proceeds of foreign export bills sent for collection, after realization overseas.

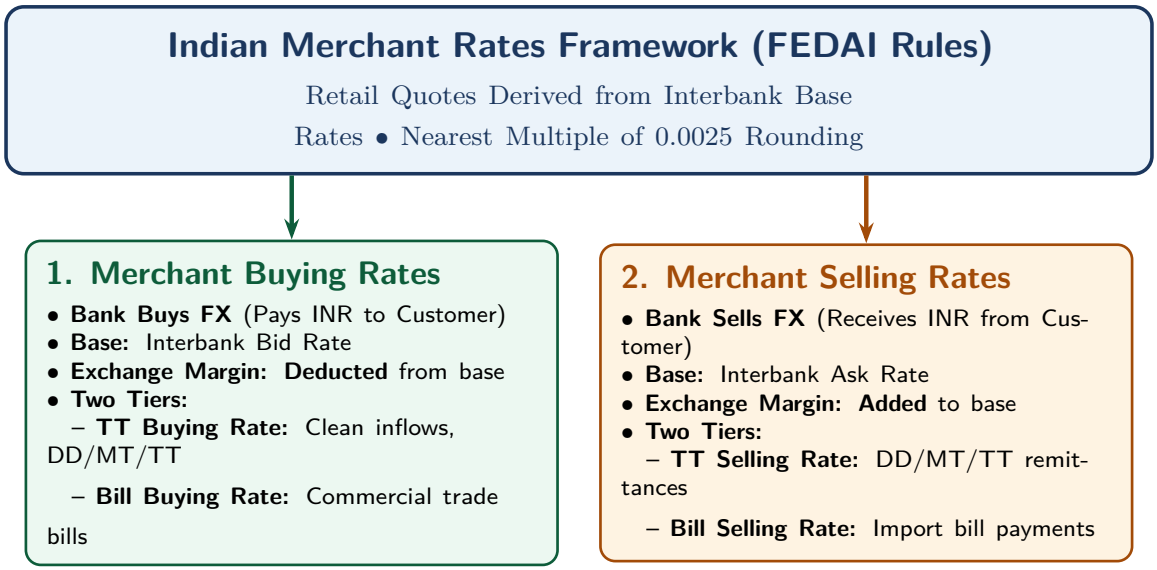


Figure 2.5: The FEDAI Merchant Exchange Rate Quotation Architecture

3. Cancellation of foreign exchange forward sale contracts entered earlier with the customer.

Quotation Formula and FEDAI Rounding Rule:

$$\text{TT Buying Rate} = \text{Interbank Spot Buying Rate (Bid)} - \text{Exchange Margin} \tag{2.7}$$

- The exchange margin is **deducted** because the bank pays fewer rupees to maximize its spread when buying foreign currency.
- **Rounding Convention:** In India, all merchant exchange rates must be rounded off to the **nearest multiple of 0.0025**.

Pro-Forma Accounting Format: TT Buying Rate Authorized dealers compute customer TT Buying quotes using the following accounting format:

Particulars	**INR**	**INR**
Dollar / Rupee market SPOT buying rate		*xxx*
Less: Exchange Margin	(*xxx*)	(*xxx*)
TT Buying Rate		**xxx**

Rounded off to nearest multiple of 0.0025 and quoted to customer.

Example 2.5.1: Calculation of TT Buying Rate (Mail Transfer)

Problem Statement:
On 15th September, you receive a Mail Transfer (MT) from your New York correspondent for 5,000 USD payable to your customer. Your account with the correspondent bank has been credited with the amount of the mail transfer. The Rupee/Dollar quotes in the local interbank market are:

$$\text{SPOT} \quad 1\,\text{USD} = \text{Rs. } 95.7600/7900$$
$$\text{SPOT / October} = 0.5600/0.5700$$

Calculate the exchange rate quoted to the customer and the rupee amount payable, given that you require an exchange margin of 0.080% (0.080 per 100) to be loaded into the rate, rounded to the nearest multiple of 0.0025.

Solution:
1. Applicable Rate Determination:
Since reimbursement has already been credited to the bank's account overseas, the applicable rate is the **TT Buying Rate**.
2. Accounting Ledger Computation:

Particulars	**INR**
Dollar / Rupee market SPOT buying rate	95.7600
Less: Exchange margin @ 0.08% on 95.7600	0.0766
Subtotal	95.6834
Rounded off to nearest multiple of 0.0025 (TT Buying Rate)	**95.6800**

3. Rupee Amount Payable to Customer:

$$\text{Rupee Amount} = 5,000\,\text{USD} \times 95.6800 = \textbf{Rs. } 4,78,400$$

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2.5.3 Bill Buying Rate Mechanics

The **Bill Buying Rate** applies when a bank purchases or discounts a foreign trade bill. Because the bank disburses funds to the domestic exporter immediately while foreign reimbursement will only occur after the normal transit period (and usance period, if applicable), the forward rate for the transit period is used:

$$\text{Bill Buying Rate} = \text{Forward Buying Rate (for Transit)} - \text{Exchange Margin} \tag{2.8}$$

Forward Margin Treatment under FEDAI Rules:

- **Foreign Currency at a Premium (Ascending Margins):** The bank gives **no benefit** of forward premium to the customer and bases the quote on the Spot Rate.
- **Foreign Currency at a Discount (Descending Margins):** The forward discount for the full transit period (or transit maturity month) is **deducted** from the spot buying rate.

Pro-Forma Accounting Format: Bill Buying Rate The standard accounting format for computing the Bill Buying Rate, incorporating transit and usance period forward adjustments, is:

Particulars	**INR**	**INR**
Dollar / Rupee market SPOT buying rate		*xxx*
ADD: Forward Premium (for transit & usance; rounded off to lower month)	*xxx*	
OR		
Less: Forward Discount (for transit & usance; rounded off to higher month)	*xxx*	(*xxx*)
Base Forward Buying Rate		*xxx*
Less: Exchange Margin		(*xxx*)
Bill Buying Rate		**xxx**

Rounded off to nearest multiple of 0.0025 and quoted to customer.

Example 2.5.2: Calculation of Bill Buying Rate (Letter of Credit)

Problem Statement:
On 25th July, your customer presents at-sight documents for 100,000 USD under an irrevocable Letter of Credit (LC). The LC provides for reimbursement by the negotiating bank's own DD on the opening bank in New York. Interbank market rates in Mumbai are:

$$\begin{aligned} \text{SPOT} \quad 1\,\text{USD} &= \text{Rs. } 95.6525/6675 \\ \text{SPOT / August} &= 0.6000/0.5700 \\ \text{SPOT / September} &= 1.0000/0.9700 \\ \text{SPOT / October} &= 1.4000/1.3900 \end{aligned}$$

The normal transit period for the bill is 25 days. Calculate the exchange rate quoted to the customer and the rupee amount payable, assuming an exchange margin of 0.15% is loaded into the rate.

Solution:
1. Transit Date and Applicable Margin:
Transaction date is 25th July. Adding 25 days of transit:

$$\text{Due Date of Reimbursement} = 25\text{th July} + 25 \text{ days} = 19\text{th August}$$

The reimbursement falls in the month of August. Because the forward margins are in descending order (0.6000/0.5700), the foreign currency is at a **discount**. The bank deducts the August forward discount (0.6000).
2. Accounting Ledger Computation:

Particulars	**INR**	**INR**
Dollar / Rupee market SPOT buying rate		95.6525
Less: Discount on 1 month (transit period 25 days)	0.6000	0.6000
Base Forward Buying Rate		95.0525
Less: Exchange margin (0.15% on Spot 95.6525)		0.1435
Net Bill Buying Rate		94.9090
Rounded off to nearest multiple of 0.0025		**94.9100**

3. Rupee Conversion Amount Payable to Customer:

$$\text{Rupee Amount} = 100{,}000\,\text{USD} \times 94.9100 = \textbf{Rs. } 94,91,000$$

2.5.4 Recovery of Interest on Bills Purchased

When commercial banks purchase or discount bills from customers, they advance rupee funds immediately. The bank claims interest from the customer covering the period from the purchase date until actual foreign realization:

$$\text{Interest on Bill} = \text{Rupee Amount} \times r \times \left[\frac{\text{NTP} + \text{Usance}}{365}\right] \tag{2.9}$$

where r is the annual bank interest rate, NTP is the Normal Transit Period in days, and Usance is the usance duration of the bill in days.

2.6 Indian Merchant Rates: Selling Rates and Mechanics

Lecture
21/09/2026

2.6.1 Nature and Types of Selling Rates

When a commercial bank sells foreign exchange, it receives INR from the customer and parts with foreign currency by issuing an international payment instrument. Authorized dealers quote two standard types of selling rates:

1. **TT Selling Rate:** Applied for clean outward remittances (issue of Demand Drafts, Mail Transfers, and Telegraphic Transfers) and cancellation of foreign exchange forward purchase contracts entered earlier.

2. **Bill Selling Rate:** Applied for processing outward commercial import bills and clean import documentary collection payments.

2.6.2 TT Selling Rate Mechanics

The TT Selling Rate is calculated on the basis of the ongoing interbank spot selling rate (Ask price):

$$\text{TT Selling Rate} = \text{Interbank Spot Selling Rate (Ask)} + \text{Exchange Margin} \tag{2.10}$$

Note: The exchange margin is **added** because the bank charges more rupees to the customer when selling foreign currency.

Pro-Forma Accounting Format: Selling Rates (TT and Bill) The FEDAI accounting format for computing merchant selling rates is:

Particulars	**INR**	**INR**
Dollar / Rupee market Spot selling rate		*xxx*
Add: Exchange margin for TT selling rate	*xxx*	*xxx*
TT Selling Rate		**xxx**
Add: Exchange margin for Bill selling rate		*xxx*
Bill Selling Rate		**xxx**

Rounded off to nearest multiple of 0.0025 and quoted to customer.

Example 2.6.1: Calculation of TT Selling Rate (Demand Draft on London)

Problem Statement:
Your customer requests you to issue a Demand Draft (DD) on London for 25,000 EUR. The ongoing rates in the local interbank market for EUR are:

$$\begin{aligned} \text{SPOT} \quad 1\,\text{EUR} &= \text{Rs. } 109.3575/3825 \\ 1 \text{ Month Forward} &= \text{Rs. } 109.7825/8250 \end{aligned}$$

Assuming an exchange margin of 0.15% is required, calculate the exchange rate quoted to the customer and the rupee amount payable by the customer.

Solution:
1. Applicable Rate Determination:
Issuing a Demand Draft requires the bank to sell foreign currency for immediate delivery. The applicable rate is the **TT Selling Rate**.

2. Accounting Ledger Computation:

Particulars	INR	INR
EUR / Rupee market Spot selling rate		109.3825
Add: Exchange margin for TT selling rate @ 0.15%	0.1641	0.1641
TT Selling Rate		**109.5466**
Rounded off to nearest multiple of 0.0025		**109.5400**

3. Total Rupee Amount Payable by Customer:

$$\text{Rupee Amount} = 25,000\,\text{EUR} \times 109.5400 = \textbf{Rs. } 27,38,500$$

2.7 Ready Rates Based on Cross Rates and Draft Cancellations

Lecture
22/09/2026

2.7.1 Principle of Cross-Rate Quotations

In the domestic Indian foreign exchange market, direct quotations are quoted for the US Dollar (USD/INR). However, exchange rates for other major currencies (such as the Canadian Dollar, Euro, Japanese Yen, or Swiss Franc) are derived by authorized dealer commercial banks based on rates prevailing in major international foreign exchange centers, including Singapore, London, Tokyo, and Hong Kong.

- **USD-Base International Quotations:** In global financial markets, currencies are predominantly quoted in terms of the US Dollar as the base or terms currency (e.g., 1 USD = 1.4551/1.4561 CAD, 1 USD = 0.8541/0.8561 EUR, 1 USD = 104.67/104.70 JPY).
- **Conversion to Indian Rupee Terms:** To quote a ready merchant rate to a domestic customer, the authorized dealer synthesizes the international USD quote with the domestic interbank USD/INR rate via cross-currency arithmetic.

2.7.2 Forward Margins and Swap Points

In international markets, forward margins are quoted as points (swap points):

- **Point Definition:** A point represents the last decimal position in the exchange quotation.
- **Four-Digit Quotations:** A point equals 0.0001. For example, a 1-month forward margin quoted as 50/60 on a spot rate of 1.5425/40 represents 0.0050/0.0060. Because margins are in ascending order ($50 < 60$), foreign currency is at a **forward premium** and margins are added to the spot base.
- **Two-Digit Quotations:** A point equals 0.01. For example, a 1-month forward margin quoted as 17/16 on a spot base of 104.67/70 represents 0.17/0.16. Because margins are in descending order ($17 > 16$), foreign currency is at a **forward discount** and margins are deducted from the spot base.

2.7.3 Mechanics of Foreign Demand Draft Cancellation

When an authorized dealer commercial bank issues a foreign currency Demand Draft (DD) on an overseas center, the transaction constitutes a **Sale of Foreign Exchange**, quoted at the prevailing **TT Selling Rate**.

If the purchaser subsequently requests cancellation of the draft and repayment of the rupee equivalent, the bank must repurchase the foreign currency from the customer. Consequently:

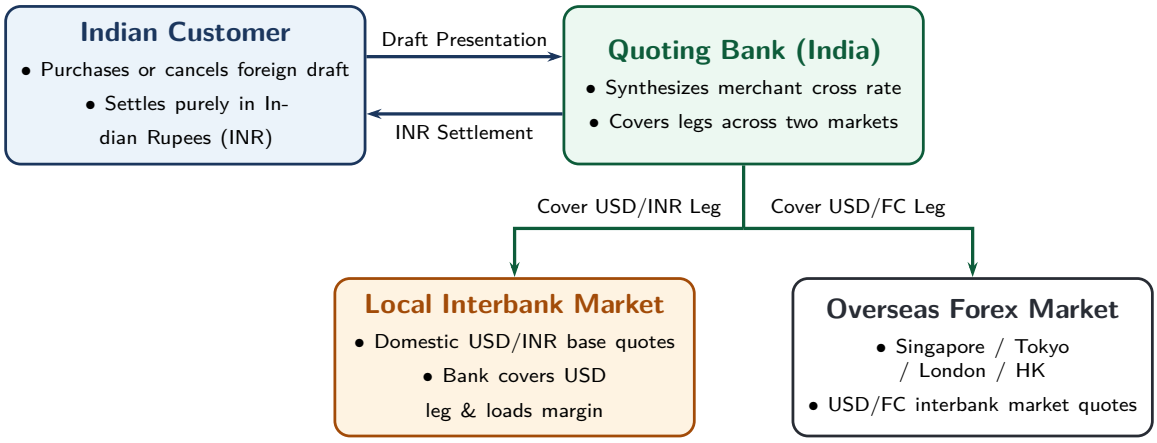


Figure 2.6: Triangular Architecture of Cross-Rate Conversion and Interbank Covering

- **Applicable Cancellation Rate:** All draft cancellations are executed strictly at the **TT Buying Rate**.
- **Cross-Rate TT Buying Formula:** To purchase foreign currency (FC) via cross rates, the bank buys USD locally at the Spot Bid rate (less exchange margin) and sells USD overseas at the market Ask (selling) rate to buy FC:

$$\text{FC TT Buying Rate} = \frac{\text{Interbank USD/INR Spot Bid} - \text{Exchange Margin}}{\text{Overseas Market USD/FC Ask Rate}} \tag{2.11}$$

- **Customer Gain or Loss on Cancellation:**

$$\text{Net Gain / Loss} = \text{Cancellation Proceeds (INR)} - \text{Original Purchase Cost (INR)} \tag{2.12}$$

A negative result represents a loss to the customer (due to merchant spreads and margin loadings), whereas a positive result indicates an exchange gain from currency appreciation.

Example 2.7.1: Cancellation of Demand Draft on Montreal (CAD 50,000)

Problem Statement:
A bank issued a Demand Draft on Montreal for CAD $50,000$ at $1\,\text{CAD} = 68.4850\,\text{INR}$. After a few days, the purchaser requested the bank to cancel the draft and repay the rupee equivalent. Assuming the Canadian Dollar was quoted in the Singapore foreign exchange market as:

$$1\,\text{USD} = 1.4551/1.4561\,\text{CAD}$$

and the ongoing quotes in the local interbank market were:

$$1\,\text{USD} = \text{Rs. } 95.5275/5350$$

Calculate the customer's gain or loss on cancellation, assuming an exchange margin of 0.08% on the TT buying rate.

Solution:
1. Applicable Rate Determination:
Cancelling the issued draft requires the bank to purchase foreign currency (CAD) back from the customer. The applicable rate is the **CAD TT Buying Rate**.

2. Accounting Ledger Computation:

Particulars	**INR**	**INR**
USD/INR Local Interbank Market Spot Buying Rate (Bid)	95.5275	
Less: Exchange Margin @ 0.08% on 95.5275 (95.5275×0.0008)	0.0764	
Net USD/INR Buying Rate		**95.4511**
USD/CAD Overseas Market Selling Rate (Ask)		1.4561
CAD TT Buying Rate (95.4511/1.4561)		**65.5526**
Rounded off to nearest multiple of 0.0025		**65.5525**

3. Gain / Loss Calculation for Customer:

Particulars	**Amount (INR)**
Amount received on cancellation ($50,000\,\text{CAD} \times 65.5525$)	32, 77, 625
Original amount paid by customer on purchase of DD ($50,000\,\text{CAD} \times 68.4850$)	34, 24, 250
Net Loss to Customer on Cancellation	**1, 46, 625**

The customer incurs a net financial loss of **Rs. 1,46,625** on cancellation.

Lecture
23/09/2026

Example 2.7.2: Cancellation of Demand Draft on London (EUR 100,000)

Problem Statement:
A bank issued a Demand Draft on London for $100,000\,\text{EUR}$ at $1\,\text{EUR} = 109.3300\,\text{INR}$. After a few days, the purchaser requested cancellation of the draft and repayment of the rupee equivalent. Assuming the Euro was quoted in the Tokyo foreign exchange market as:

$$1\,\text{USD} = 0.8541/0.8561\,\text{EUR}$$

and the ongoing rates in the local interbank market were:

$$1\,\text{USD} = \text{Rs. } 95.6700/6720$$

Calculate the customer's gain or loss on cancellation, assuming an exchange margin of 0.08% on the TT buying rate.

Solution:
1. Applicable Rate Determination:
The bank repurchases the Euro draft from the customer, applying the **EUR TT Buying Rate**.
2. Accounting Ledger Computation:

Particulars	INR	INR
USD/INR Local Interbank Market Spot Buying Rate (Bid)	95.6700	
Less: Exchange Margin @ 0.08% on 95.6700 (95.6700×0.0008)	0.0765	
Net USD/INR Buying Rate		**95.5935**
USD/EUR Overseas Market Selling Rate (Ask)		0.8561
EUR TT Buying Rate (95.5935/0.8561)		**111.6616**
Quoted Rate (Preserved per Note Record)		**111.6616**

3. Gain / Loss Calculation for Customer:

Particulars	Amount (INR)
Amount received on cancellation ($100,000\,\text{EUR} \times 111.6616$)	$1,11,66,160$
Original amount paid by customer on purchase ($100,000\,\text{EUR} \times 109.3300$)	$1,09,33,000$
Net Gain to Customer on Cancellation	$\mathbf{2,33,160}$

Owing to the substantial market appreciation of the Euro against the Rupee between the issue and cancellation dates, the customer realizes a net gain of **Rs. 2,33,160**.

2.8 Forward Exchange Contracts and FEDAI Rule 7

2.8.1 Concept and Economic Purpose of Forward Contracts

Lecture
24/09/2026

A **Forward Exchange Contract (FEC)** is an operational risk management instrument designed to provide complete protection to an importer or exporter against adverse fluctuations in foreign exchange rates (exchange rate risk).

Definition 2.8.1: Forward Exchange Contract (FEC)

A **Forward Exchange Contract** is a legally binding agreement entered into between an authorized dealer bank and a customer (or between two commercial banks in the interbank tier) to buy or sell a specified amount of foreign currency at a predetermined exchange rate on an agreed future date or within an agreed future period.

By booking an FEC, commercial counterparties lock in their conversion rates at inception:

- **Exporters (Forward Sale / Bank Purchases):** Eliminate the risk of foreign currency depreciation, fixing the exact rupee cash realization upon receipt of export proceeds.
- **Importers (Forward Purchase / Bank Sells):** Eliminate the risk of foreign currency appreciation, establishing the exact rupee expenditure required to settle overseas import payables.

2.8.2 Delivery Date Computation under FEDAI Rule 7

A fundamental statutory standard established by FEDAI (Foreign Exchange Dealers' Association of India) governs the contractual maturity and delivery date of all forward exchange contracts booked in India:

FEDAI Rule 7: Commencement of Forward Duration

Under **FEDAI Rule 7**, the delivery date of a forward contract is deliverable at a future date where the contract duration is computed strictly from the **Spot Value Date** $(T + 2)$ of the transaction, and **NOT** from the trade booking date (T).

Operational Example: If a customer books a 3-month forward exchange contract on **12th February**:

1. The trade date is $T = 12$th February.
2. The spot value date is $T + 2 = 14$th February (two business days post-trade).
3. The 3-month contractual tenor commences on **14th February**, and the forward contract matures exactly on **14th May**.

2.8.3 Classification: Fixed vs. Option Forward Contracts

Depending on the flexibility of the settlement date, forward exchange contracts are categorized into two structural types:

1. **Fixed Forward Contract:** A forward contract where the delivery and physical settlement of foreign exchange must take place strictly on a **specified future calendar date**.
   - *Example:* On 5th March, a customer enters into a 3-month fixed forward sale contract for $10,000$ EUR. Computing from the spot date (7th March), the contract matures on **7th June**. The customer must present the underlying bill or foreign exchange instrument to the bank exactly on 7th June. Delivery cannot occur prior to or later than this designated date.
2. **Option Forward Contract:** A forward contract where the customer possesses the contractual option to deliver or take delivery of foreign exchange on **any business day during a specified contractual window** at the predetermined exchange rate.
   - *Example:* On 15th September, a customer enters into a 2-month forward contract with "option over November". The customer is entitled to deliver foreign exchange to the bank on any working day between **1st November and 30th November**.

FEDAI Rule 7 Statutory Option Window Constraint: Under FEDAI Rule 7, the option delivery period granted to the customer shall **not exceed one calendar month**. Option contracts cannot span indefinite or multi-month settlement windows.

2.8.4 Statutory Terms and Conditions of Forward Contracts

All forward exchange contracts entered into by authorized dealers in India are governed by standard terms and conditions prescribed under FEDAI rules:

1. **Commercial Backing:** Every forward exchange contract booked by a customer must be supported by genuine, underlying foreign trade exposure (firm export order, import letter of credit, or authorized external commercial borrowing).

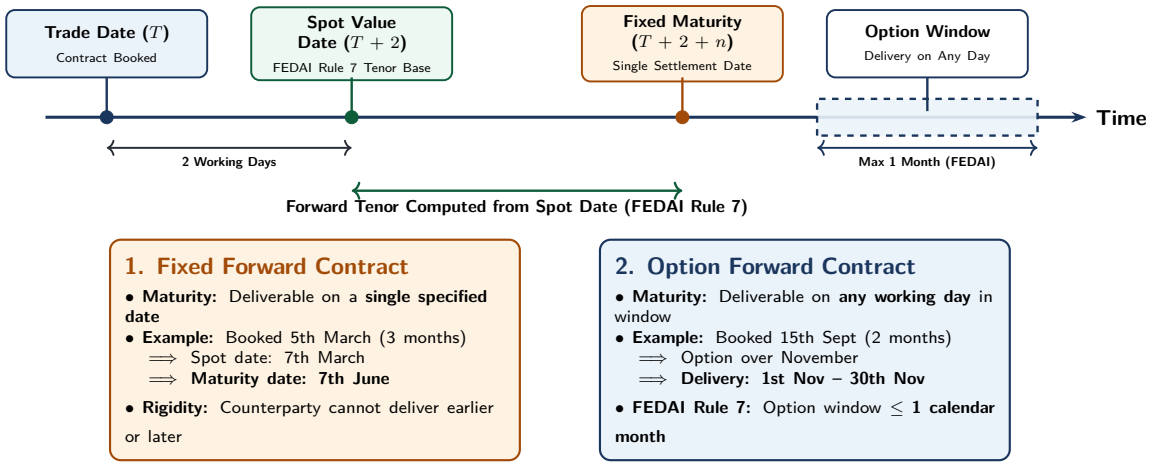


Figure 2.7: Chronological Delivery Horizon Architecture under FEDAI Rule 7

2. **Contractual Details:** The contract confirmation note must unambiguously state the contract date, counterparty name, currency, contract amount, agreed forward exchange rate, spot value baseline, and exact delivery horizon.

3. **Customer Margin Requirements:** Banks may stipulate an initial cash margin or credit line to protect against potential default risk arising from adverse currency movements prior to settlement.

4. **Non-Performance and Cancellation:** If the customer fails to deliver foreign exchange on the maturity date (or by the end of the option delivery window), the contract must be automatically cancelled by the bank on the 15th day post-maturity (or as prescribed by prevailing FEDAI rules) at the ongoing merchant buying/selling rate, with any resulting exchange loss recovered from the customer.

2.9 Forward Merchant Rates and Delivery Option Mechanics

2.9.1 Mechanics of Forward Merchant Bill Buying Rates

Lecture
08/10/2026

When an authorized dealer commercial bank enters into a forward purchase contract for an export bill, the date on which foreign exchange proceeds are physically realized depends on the classification of the underlying instrument:

- **Demand Bill:** Foreign currency funds are realized post-shipment after the **Normal Transit Period (NTP)**, standardly stipulated under FEDAI rules as **25 days**.
- **Usance Bill:** Realization occurs only upon maturity of the credit period, encompassing both the **Normal Transit Period** and the agreed **Usance Period** (e.g., 25 days transit + 60 days usance = 85 days).

Consequently, when quoting a forward merchant bill buying rate, the total duration for which the quoting bank must cover itself in the interbank market is:

$$\text{Total Interbank Cover Tenor} = \text{Forward Period} + \text{Transit Period} + \text{Usance Period} \tag{2.13}$$

Statutory FEDAI Rules for Forward Swap Margins

Because interbank forward margins are quoted on a monthly basis, fractional holding periods (in days) must be rounded to an integer number of months. To ensure the quoting bank remains fully hedged and does not grant undue speculative gain to the merchant:

1. **Ascending Swap Margins (Foreign Currency at Forward Premium):** Adding a premium increases the buying rate paid to the exporter. Under FEDAI guidelines, the bank must give the minimum benefit to the customer by **rounding down** the total duration to the **lower whole month** (e.g., 25 days transit → 0 months; 85 days → 2 months; 145 days → 4 months).
2. **Descending Swap Margins (Foreign Currency at Forward Discount):** Deducting a discount lowers the buying rate. To ensure the bank is adequately protected against market depreciation, the bank must **round up** the total duration to the **higher whole month**.
3. **Exchange Margin Deduction:** The standard merchant loading percentage (e.g., 0.10%) is computed on the base buying rate and deducted.
4. **FEDAI Rounding Rule:** The resulting merchant quote is rounded off to the nearest multiple of 0.0025.

Example 2.9.1: Multi-Tier Forward Bill Buying Rate Determinations

Problem Statement:
An authorized dealer bank is requested to quote merchant buying rates for US Dollars under the following ongoing interbank market quotations:

Tenor	Bid Rate	Ask Rate
Spot USD/INR	96.6000	96.6075
1-Month Forward Margin	0.3500	0.3600
2-Month Forward Margin	0.5500	0.5600
3-Month Forward Margin	0.8500	0.8600
4-Month Forward Margin	1.1500	1.1600
5-Month Forward Margin	1.3500	1.3600
6-Month Forward Margin	1.5500	1.6600

The normal transit period is 25 days. All forward contracts are for fixed delivery. The bank requires an exchange margin of 0.10%. Calculate: (a) Ready bill buying rate, (b) 2 months forward buying rate for a demand bill, (c) Ready rate for a 60 days usance bill, and (d) 2 months forward buying rate for a 60 days usance bill.

Solution:

(a) Computation of Ready Bill Buying Rate:
For a ready demand bill, the holding period is the 25-day transit period. Since margins are ascending (premium), 25 days rounds down to 0 months, so no forward premium is added:

Particulars	**INR**	**INR**
USD/INR Market Spot Buying Rate (Bid)		96.6000
Less: Exchange Margin @ 0.10% on 96.6000 (96.6000 × 0.0010)		0.0966
Subtotal (before rounding)		**96.5034**
Quoted Ready Bill Buying Rate (Rounded to nearest 0.0025)		**96.5025**

(b) Computation of 2 Months Forward Buying Rate for Demand Bill:
Total duration = Forward Period (2 months) + Transit Period (25 days). Under premium, 25 days rounds down to 0 months, making the applicable forward tenor 2 months (premium = 0.5000 per bank ledger):

Particulars	**INR**	**INR**
USD/INR Market Spot Buying Rate (Bid)		96.6000
Add: Forward Premium for 2 Months (Transit 25 days rounded to lower month)		0.5000
Gross Forward Buying Rate		97.1000
Less: Exchange Margin @ 0.10% on 97.1000 (97.1000 × 0.0010)		0.0971
Subtotal (before rounding)		**97.0029**
Quoted 2-Month Forward Demand Bill Rate (Rounded to nearest 0.0025)		**97.0000**

(c) Computation of Ready Rate for 60 Days Usance Bill:
Total duration = Transit Period (25 days) + Usance Period (60 days) = 85 days. Under premium, 85 days rounds down to 2 months (applicable 2-month bid premium = 0.5500):

Particulars	**INR**	**INR**
USD/INR Market Spot Buying Rate (Bid)		96.6000
Add: Forward Premium for 2 Months (25d transit + 60d usance = 85d → 2m)		0.5500
Gross Buying Rate		97.1500
Less: Exchange Margin @ 0.10% on 97.1500 (97.1500 × 0.0010)		0.09715
Subtotal (before rounding)		**97.05285**
Quoted Ready 60-Day Usance Bill Rate (Rounded to nearest 0.0025)		**97.0525**

(d) Computation of 2 Months Forward Rate for 60 Days Usance Bill:
Total duration = Forward Period (2 months) + Transit (25 days) + Usance (60 days) = 145 days. Under premium, 145 days rounds down to 4 months (applicable 4-month bid premium = 1.1500):

Particulars	**INR**	**INR**
USD/INR Market Spot Buying Rate (Bid)		96.6000
Add: Forward Premium for 4 Months (2m forward+25d transit+ 60d usance → 4m)		1.1500
Gross Forward Buying Rate		97.7500
Less: Exchange Margin @ 0.10% on 97.7500 (97.7500 × 0.0010)		0.09775
Subtotal (before rounding)		**97.65225**
Quoted 2-Month Forward 60-Day Usance Rate (Rounded to nearest 0.0025)		**97.6525**

2.9.2 Option Forward Contracts and Delivery Window Margin Rules

Lecture
12/10/2026

Under FEDAI Rule 7, an option forward contract provides the customer with operational flexibility to execute delivery on any working day within an agreed calendar window (not exceeding one month).

Because the exact date of settlement is elected exclusively by the merchant, the quoting bank faces uncertainty regarding the exact interbank swap points to apply. To eliminate counterparty

selection risk, FEDAI establishes the **Worst-Of Principle**:

FEDAI "Worst-Of" Principle for Option Contracts

The quoting bank must always compute the merchant forward rate on the assumption that the customer will elect to deliver foreign exchange on the specific calendar date within the option window that is **least advantageous to the bank** (and most favorable to the customer).

Statutory Delivery Date Selection Protocol

The operational date assumed by the bank depends on whether the transaction is a bank purchase or bank sale, and whether the currency is at a premium or discount:

1. **Forward Purchase by Bank (Customer Sells Foreign Exchange):**
   - *Foreign Currency at Premium:* Adding premium increases the payout to the customer. Therefore, the bank quotes the **Earliest Delivery Date** of the option window, giving the customer the lowest premium.
   - *Foreign Currency at Discount:* Deducting discount reduces the payout. Therefore, the bank quotes the **Latest Delivery Date** of the option window, deducting the maximum discount.
2. **Forward Sale by Bank (Customer Buys Foreign Exchange):**
   - *Foreign Currency at Premium:* Adding premium increases the cost to the customer. Therefore, the bank quotes the **Latest Delivery Date** of the option window, charging the maximum premium.
   - *Foreign Currency at Discount:* Deducting discount reduces the cost. Therefore, the bank quotes the **Earliest Delivery Date** of the option window, granting the minimum discount.

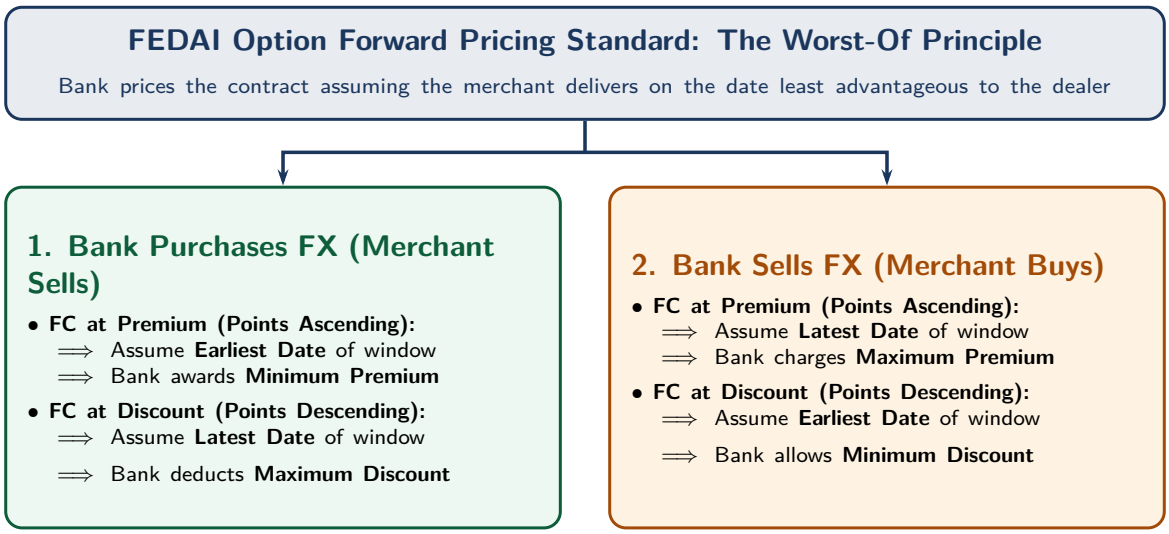


Figure 2.8: FEDAI Statutory Margin Selection Architecture for Option Forward Contracts

Example 2.9.2: Option Forward Contract for Export Bill on New York

Problem Statement:
On 15th July, an export customer requests an authorized dealer bank to book a forward exchange contract with delivery during September covering a 30-day sight bill drawn on New York for USD 65,000 under an irrevocable Letter of Credit.
Assuming US Dollars are quoted in the local interbank market on Spot August 2026 baseline as follows:

Tenor	Bid Rate	Ask Rate
Spot USD/INR	96.3700	96.3800
Spot / August Forward Margin	0.0300	0.0400
Spot / September Forward Margin	0.1100	0.1300
Spot / October Forward Margin	0.1900	0.2200
Spot / November Forward Margin	0.2700	0.3100
Spot / December Forward Margin	0.3500	0.4000

Assuming the bank charges an exchange margin of 0.10% on the spot buying rate and the normal transit period is 25 days, calculate: (i) the forward bill buying rate quoted to the customer, and (ii) the total rupee proceeds payable upon realization.

Solution:

1. Transaction Characterization and Rate Determination:

- **Contract Type:** Forward Purchase of an Export Bill (Bank purchases USD).
- **Delivery Option Window:** 1st September to 30th September.
- **Margin Structure:** Forward points are ascending ($0.0300 < 0.1100 < 0.1900$), indicating USD is at a **Forward Premium**.
- **Worst-Of Rule Application:** Under forward premium, the bank assumes the customer delivers on the **earliest possible date** in the window (1st September) to award the minimum premium, applying the September bid margin (0.1100).

2. Accounting Ledger Computation:

Particulars	**INR**	**INR**
USD/INR Market Spot Buying Rate (Bid)		96.3700
Add: Forward Premium for September (Bid Side)		0.1100
Gross Forward Buying Rate		96.4800
Less: Exchange Margin @ 0.10% on Spot Buying Rate (96.3700×0.0010)		0.0964
Bill Buying Rate (before rounding)		**96.3836**
Quoted Rate (Rounded off to nearest multiple of 0.0025)		**96.3800**

3. Total Rupee Realization:

$$\text{Rupee Amount Payable} = \text{USD } 65{,}000 \times 96.3800 = \text{Rs. } \mathbf{62{,}64{,}700} \tag{2.14}$$

The customer receives exactly Rs. **62,64,700** upon delivering the export bill.

2.10 Forward Exchange Rates Based on Cross Rates

Lecture
14/10/2026

2.10.1 Principles of Cross-Rate Forward Quotations

When an Indian exporter or importer requires forward exchange cover in a non-USD currency (e.g., SGD, JPY, EUR, GBP, CHF), the quotation cannot be established directly from domestic rupee interbank quotes because non-dollar currencies are not actively traded against the Rupee in the domestic interbank market.

Instead, commercial authorized dealers synthesize the required forward merchant rate via **Triangular Interbank Covering**, entering into two concurrent wholesale hedging transactions:

1. **Domestic Interbank Tier (Mumbai):** An interbank USD/INR forward contract for the underlying delivery month.
2. **Overseas Interbank Tier (Singapore/Tokyo/London):** An interbank forward contract in USD against the foreign currency (USD/FC) for the required duration.

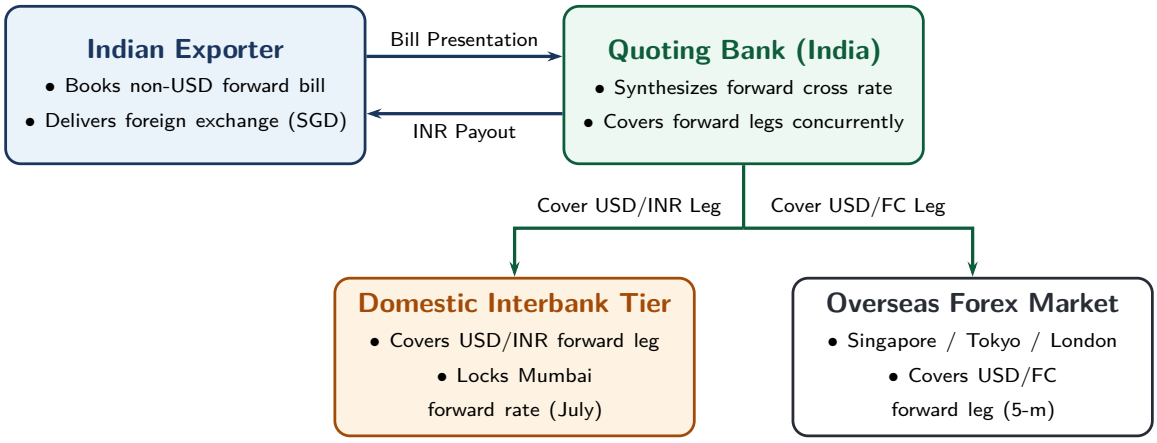


Figure 2.9: Dual-Leg Wholesale Covering Architecture for Forward Cross-Rate Contracts

2.10.2 Textbook Pro-Forma Quotation Formats

FEDAI guidelines establish standardized accounting statement templates for computing forward merchant rates on a cross-rate basis:

Format 9.1: Calculation of Forward Buying Rate for Other Currencies		
Dollar/Rupee Market Spot Buying Rate	= Rs.	
Add: Forward Premium (for forward, transit, & usance: rounded to lower month)		
OR *Less:* Forward Discount (rounded to higher month)	± Rs.	
Subtotal	= Rs.	
Less: Exchange Margin	−Rs.	
Forward Buying Rate for Dollar	= Rs. .....*	[**1**]
Dollar/Foreign Currency Market Spot Selling Rate	= FC	
Add: Forward Premium (for forward, transit, & usance: rounded to higher month)		
OR *Less:* Forward Discount (rounded to lower month)	± FC	
Forward Selling Rate for Foreign Currency	= FC	[**2**]
Forward Buying Rate for Foreign Currency = [1] ÷ [2]	= Rs. .....*	[**3**]

*Rounded off to the nearest multiple of 0.0025 when quoted to the customer.

Format 9.2: Calculation of Forward Selling Rate for Other Currencies		
Dollar/Rupee Market Spot Selling Rate	= Rs.	
Add: Forward Premium (for forward period)		
OR *Less:* Forward Discount (for forward period)	± Rs.	
Subtotal	= Rs.	
Add: Exchange Margin to TT Selling Rate	+Rs.	
Forward TT Selling Rate for Dollar	= Rs. .....*	[**1**]
Add: Exchange Margin for Bill Selling Rate	+Rs.	
Forward Bill Selling Rate for Dollar	= Rs. .....*	[**2**]
Dollar/Foreign Currency Spot Buying Rate	= FC	
Add: Forward Premium (for forward period)		
OR *Less:* Forward Discount (for forward period)	± FC	
Forward Buying Rate for Foreign Currency	= FC	[**3**]
Forward TT Selling Rate for Foreign Currency = [1] ÷ [3]	= Rs. .....*	[**4**]
Forward Bill Selling Rate for Foreign Currency = [2] ÷ [3]	= Rs. .....*	[**5**]

*Rounded off to the nearest multiple of 0.0025 when quoted to the customer.

Example 2.10.1: Forward Purchase of Export Bill in Singapore Dollars

Problem Statement:
An export customer requests an authorized dealer bank to book a forward contract for an export bill of SGD 1,00,000. The bill is drawn under an irrevocable Letter of Credit with a usance period of 4 months and 5 days sight. The normal transit period is 25 days.
The ongoing interbank market rates are quoted as follows:

$$\begin{aligned}\text{USD/INR Spot} &= 96.4875/4925\\ \text{USD/INR Spot / July Forward Margin} &= 0.4600/0.4800\\ \text{USD/SGD Spot} &= 1.4004/1.4078\\ \text{USD/SGD 5-Month Forward Margin} &= 0.0235/0.0265\end{aligned}$$

Assuming an exchange margin of 0.10% is required, calculate the forward exchange rate quoted to the customer and the total rupee amount payable, given that the bank covers the USD/INR leg for July and the USD/SGD leg for 5 months.

Solution:

1. Total Duration Determination:

$$\begin{aligned}\text{Total Duration} &= \text{Usance Period (4 months 5 days)} + \text{Transit Period (25 days)}\\ &= 4 \text{ months } 30 \text{ days} = \mathbf{5} \text{ months}\end{aligned}$$

The bank covers the USD/INR leg in the Mumbai market for July (premium = 0.4600) and the USD/SGD leg in the Singapore market for 5 months (premium = 0.0235).

2. Accounting Ledger Computation:

Particulars	**INR**	**SGD**
Stage 1: Domestic Interbank Dollar Buying Rate		
USD/INR Spot Buying Rate (Bid)	96.4875	
Add: Forward Premium for July	0.4600	
Gross Forward Buying Rate for Dollar	96.9475	
Less: Exchange Margin @ 0.10% on 96.9475 (96.9475×0.0010)	0.0970	
Net Forward Buying Rate for Dollar [1]	**96.8505**	
Stage 2: Overseas Interbank Dollar Selling Rate		
USD/SGD Spot Selling Rate (Ask)		1.4078
Add: Forward Premium for 5 Months		0.0235
Forward Selling Rate for Dollar against SGD [2]		**1.4313**
Stage 3: Synthesized SGD Forward Buying Rate		
Forward Buying Rate for SGD = [1] ÷ [2] = $\frac{96.8505}{1.4313}$		**67.6661**
Quoted Rate (Rounded off to nearest multiple of 0.0025)		**67.6625**

3. Total Rupee Amount Payable to Customer:

$$\text{Rupee Amount Payable} = \text{SGD } 1{,}00{,}000 \times 67.6625 = \text{Rs. } \mathbf{67{,}66{,}250} \tag{2.15}$$

The customer receives Rs. **67,66,250** upon realization of the Singapore Dollar export bill.

2.11 Theories of Foreign Exchange Rate Determination

Lecture
21/10/2026

Understanding the macroeconomic forces that dictate exchange rate movements is essential for corporate treasury operations, cross-border pricing, and international investment strategy. Exchange rate determination models formalize the fundamental relationships linking money supply growth, price level inflation, interest rate differentials, and forward foreign exchange margins.

2.11.1 The Law of One Price and Purchasing Power Parity (PPP)

The foundational premise of foreign exchange valuation is the **Law of One Price (LOOP)**. It postulates that in frictionless international markets—absent transportation costs, import tariffs, and trade barriers—an identical basket of tradable goods must sell for the exact same price across different countries when converted into a common currency:

$$P_d = S \times P_f \quad \Longleftrightarrow \quad S = \frac{P_d}{P_f} \tag{2.16}$$

where P_d represents the domestic currency price, P_f is the foreign currency price, and S is the spot exchange rate quoted in direct terms (domestic currency units per unit of foreign currency).

Classroom Commodity Basket Illustration

Consider a standardized consumer commodity (e.g., 1 kg of oranges) whose market price is observed in India and the United States:

- At baseline $t = 0$, 1 kg of oranges costs Rs. 60 in India and \$1.00 in the US.
- Under the Law of One Price, the implied equilibrium exchange rate is:

$$\text{Rs. } 60 = 1\,\text{kg orange} = \$1.00 \implies S_0 = \text{Rs. } 60\,/\$$$

When inflation rates diverge between the two economies, the purchasing power of money erodes unevenly. Suppose an extended consumption basket initially costs Rs. 9,000 in India and \$100 in the United States, establishing the spot rate $S_0 = \frac{9{,}000}{100} = \text{Rs. } 90\,/\$$. Over an annual period, the annual rate of inflation is 8% in India and 5% in the United States:

$$P_{d,1} = \text{Rs. } 9{,}000 \times (1 + 0.08) = \text{Rs. } 9{,}720$$
$$P_{f,1} = \$100 \times (1 + 0.05) = \$105$$

To restore purchasing power equivalence, the revised exchange rate S_1 must adjust to:

$$S_1 = \frac{\text{Rs. } 9{,}720}{\$105} = \text{Rs. } \mathbf{92.57}\,/\$$$

The Indian Rupee depreciates by Rs. 92.57 − Rs. 90.00 = Rs. 2.57 /\$, exactly offsetting the 3% inflation differential (8% − 5%) between the two trading partners.

Absolute vs. Relative Purchasing Power Parity

Economists distinguish between two formal variants of PPP:

1. **Absolute Purchasing Power Parity:** Requires that the spot exchange rate directly equals the ratio of the overall national price level indexes:

$$S_t = \frac{P_{d,t}}{P_{f,t}} \tag{2.17}$$

   Absolute PPP rarely holds strictly in empirical reality due to transport costs, tariffs, non-traded services, and differing consumer preferences across countries.

2. **Relative Purchasing Power Parity:** Relaxes the requirement of identical absolute price levels and instead asserts that the *percentage change* in the spot exchange rate over any time horizon t is determined by the inflation differential between the domestic and foreign economies:

$$S_t = S_0 \left(\frac{1 + i_L}{1 + i_F} \right)^t \tag{2.18}$$

where i_L is the domestic (local) inflation rate, i_F is the foreign inflation rate, and t is the time horizon in years. If the domestic country experiences higher inflation than the foreign partner ($i_L > i_F$), the domestic currency must depreciate relative to the foreign currency to preserve export competitiveness.

2.11.2 Macroeconomic Parity Architecture

Lecture 22/10/2026

The interaction between the money supply, price inflation, nominal interest rates, and foreign exchange rates forms a closed, mutually reinforcing equilibrium system often called the **International Parity Web**.

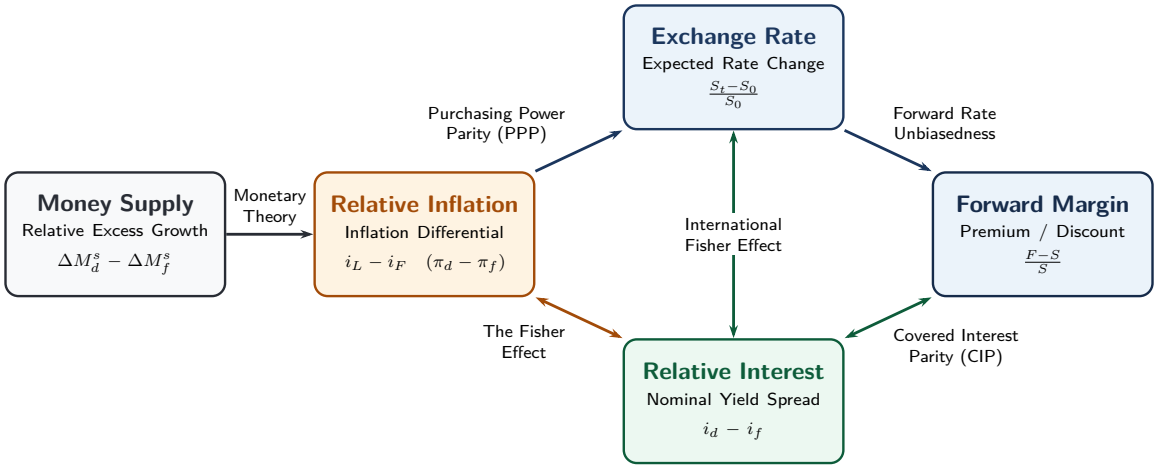


Figure 2.10: Macroeconomic Parity Architecture: Interrelationships between Money, Inflation, Interest Rates, and Exchange Rates

As mapped in Figure 2.10, the international transmission chain proceeds as follows:

1. **Monetary Theory of Inflation:** When the domestic supply of money expands at a rate faster than real output demand, the purchasing power of the currency declines, generating generalized price inflation ($\Delta M^s \rightarrow \Delta \pi$).

2. **Purchasing Power Parity (PPP):** Cross-border goods arbitrage translates higher relative inflation into an expected currency depreciation ($\Delta \pi \rightarrow \Delta S$).

3. **The Fisher Effect:** Lenders demand higher nominal interest rates to compensate for anticipated inflation, driving the domestic nominal yield above foreign yields ($\Delta \pi \leftrightarrow \Delta i$).

4. **International Fisher Effect (IFE):** Capital flows equalize expected returns across borders, ensuring that nominal interest rate differentials reflect expected exchange rate movements ($\Delta S \leftrightarrow \Delta i$).

5. **Forward Rate Unbiasedness:** Under market efficiency, the forward exchange rate serves as an unbiased predictor of the future spot rate ($\Delta S \rightarrow \frac{F-S}{S}$).

6. **Covered Interest Parity (CIP):** Covered interest arbitrage eliminates riskless arbitrage between money market rate differentials and forward exchange rate swap points ($\Delta i \leftrightarrow \frac{F-S}{S}$).

2.11.3 The Fisher Effect and Monetary Inflation Dynamics

The **Fisher Effect** establishes that nominal interest rates in any economy encompass two distinct components: a real rate of return and an inflation compensation premium:

$$1 + r_{\text{nominal}} = (1 + r_{\text{real}})(1 + i_{\text{inflation}}) \tag{2.19}$$

Expanding this identity yields:

$$r_{\text{nominal}} = r_{\text{real}} + i_{\text{inflation}} + (r_{\text{real}} \times i_{\text{inflation}})$$

Because the product term $(r_{\text{real}} \times i_{\text{inflation}})$ is negligible under moderate inflation environments, the classical linear Fisher approximation is widely applied:

$$r_{\text{nominal}} \approx r_{\text{real}} + i_{\text{inflation}} \tag{2.20}$$

Under integrated global capital markets with free financial mobility, real rates of return tend toward international convergence ($r_{\text{real},d} \approx r_{\text{real},f}$). Consequently, the nominal interest rate differential between two countries directly reflects their expected inflation differential:

$$r_d - r_f \approx \pi_d - \pi_f \tag{2.21}$$

2.11.4 Worked Examples on Purchasing Power Parity

Example 2.11.1: One-Year and Two-Year Exchange Rate Forecasting via Relative PPP

Problem Statement:
The expected annual rate of inflation in India is 5% and in the United States is 3%. If the initial spot exchange rate is Rs. 97 /\$, calculate the expected exchange rate:

(a) After one year ($t = 1$), and
(b) After two years ($t = 2$).

Solution:
1. Theoretical Formulation:
Under the Relative Purchasing Power Parity framework:

$$S_t = S_0 \left(\frac{1 + i_L}{1 + i_F} \right)^t$$

Given:

- Base Spot Rate $S_0 = \text{Rs. } 97.00 \text{ /\$}$
- Domestic (India) Inflation $i_L = 5\% = 0.05$
- Foreign (USA) Inflation $i_F = 3\% = 0.03$
- Relative Inflation Factor $= \frac{1+0.05}{1+0.03} = \frac{1.05}{1.03} \approx 1.019417$

2. Step-by-Step Forecast Calculation:

(a) **Exchange Rate after One Year ($t = 1$):**

$$\begin{aligned} S_1 &= 97.00 \times \left(\frac{1.05}{1.03} \right)^1 = 97.00 \times 1.019417 \\ &= \text{Rs. } \mathbf{98.88} \text{ /\$} \end{aligned}$$

The Indian Rupee depreciates against the US Dollar to Rs. 98.88 /\$.

(b) **Exchange Rate after Two Years ($t = 2$):**

$$\begin{aligned} S_2 &= 97.00 \times \left(\frac{1.05}{1.03} \right)^2 = 97.00 \times (1.019417)^2 \\ &= 97.00 \times 1.039212 = \text{Rs. } \mathbf{100.80} \text{ /\$} \end{aligned}$$

Alternatively, compounding from year 1: $S_2 = 98.8835 \times 1.019417 = \text{Rs. } \mathbf{100.80}$ /\$.

Example 2.11.2: Multi-Year Term-Structure Exchange Rate Projection under Relative PPP

Problem Statement:
The prevailing spot exchange rate is \$1.65 /GBP. The projected inflation rates in the United Kingdom and the United States for the next three years are structured as follows:

Year	UK Expected Inflation (i_{UK})	US Expected Inflation (i_{US})
1	3.0%	2.0%
2	3.5%	2.5%
3	3.0%	2.0%

Calculate the expected spot exchange rate (\$/GBP) at the end of Year 1, Year 2, and Year 3.

Solution:
1. Model Specification:
Here, the quote is expressed in American terms (US Dollars per unit of British Pound, \$ /GBP). The British Pound (GBP) is the base currency and the US Dollar (USD) is the terms (pricing) currency. According to Relative PPP:

$$S_t = S_{t-1} \times \left(\frac{1 + i_{\text{US},t}}{1 + i_{\text{UK},t}} \right)$$

2. Sequential Year-by-Year Compounding:

- **End of Year 1:**

$$S_1 = 1.6500 \times \frac{1 + 0.020}{1 + 0.030} = 1.6500 \times \frac{1.020}{1.030} = 1.6500 \times 0.990291 = \mathbf{\$1.6340}\,/\text{GBP}$$

- **End of Year 2:**

$$S_2 = 1.63398 \times \frac{1 + 0.025}{1 + 0.035} = 1.63398 \times \frac{1.025}{1.035} = 1.63398 \times 0.990338 = \mathbf{\$1.6182}\,/\text{GBP}$$

- **End of Year 3:**

$$S_3 = 1.61819 \times \frac{1 + 0.020}{1 + 0.030} = 1.61819 \times \frac{1.020}{1.030} = 1.61819 \times 0.990291 = \mathbf{\$1.6025}\,/\text{GBP}$$

3. Cumulative Term-Structure Projection Ledger:

Year (t)	UK Inflation (i_{UK})	US Inflation (i_{US})	PPP Multiplier $\left(\frac{1+i_{\text{US}}}{1+i_{\text{UK}}}\right)$	Projected Spot Rate (S_t)
0	—	—	Baseline Spot	\$1.6500 /GBP
1	3.0%	2.0%	$\frac{1.020}{1.030} \approx 0.990291$	**\$1.6340** /GBP
2	3.5%	2.5%	$\frac{1.025}{1.035} \approx 0.990338$	**\$1.6182** /GBP
3	3.0%	2.0%	$\frac{1.020}{1.030} \approx 0.990291$	**\$1.6025** /GBP

Because the United Kingdom consistently exhibits higher inflation than the United States in all three years, the British Pound depreciates monotonically from \$1.6500 /GBP to \$1.6025 /GBP.

2.11.5 Interest Rate Parity (IRP) and Covered Interest Arbitrage

Lecture
26/10/2026

While Purchasing Power Parity connects goods markets through commodity arbitrage over medium-to-long horizons, **Interest Rate Parity (IRP)** governs financial markets through money market arbitrage over short-to-medium horizons.

Benchmark Spot and Forward Outright Quotations

In foreign exchange operations:

- **Spot Rate (S):** The outright benchmark rate for immediate delivery ($T + 2$ business days).
- **Forward Rate (F):** The contracted rate established today for delivery and settlement at a designated future maturity date ($T > 2$).

Under market equilibrium, the percentage forward premium or discount on a foreign currency equals the nominal interest rate differential between the domestic and foreign financial centers:

$$F = S \times \left(\frac{1 + i_d}{1 + i_f} \right) \tag{2.22}$$

where i_d is the domestic interest rate and i_f is the foreign interest rate for the matching tenor. Subtracting S from both sides yields the classical forward premium relationship:

$$\frac{F - S}{S} = \frac{i_d - i_f}{1 + i_f} \approx i_d - i_f \tag{2.23}$$

The currency associated with the higher nominal interest rate trades at a forward discount, while the currency with the lower nominal interest rate trades at a forward premium.

Covered Interest Arbitrage Mechanics

When market forward rates deviate from the theoretical parity condition defined by Figure 2.10, an arbitrageur can execute a four-stage riskless transaction sequence without bearing foreign exchange risk:

Example 2.11.3: Covered Interest Arbitrage Execution and CIP Verification

Problem Statement:
An arbitrageur observes 1-year US interest rate $i_{\text{US}} = 5.0\%$, UK interest rate $i_{\text{UK}} = 8.0\%$, spot rate $S = \$1.50\,/\text{GBP}$, and 1-year forward rate $F = \$1.48\,/\text{GBP}$. Demonstrate how borrowing \$1,000,000 in the US to execute covered interest arbitrage yields a net riskless arbitrage profit, and contrast the market forward rate against the theoretical Covered Interest Parity (CIP) benchmark.

Solution:

1. Four-Stage Covered Interest Arbitrage Sequence:

- **Stage 1 (Domestic Borrowing & Spot Conversion, $t = 0$):** Borrow \$1,000,000 in the US at 5% and convert to Sterling at $S = \$1.50\,/\text{GBP} \implies$ Sterling Proceeds $= \frac{\$1{,}000{,}000}{1.50} = £\,\mathbf{666{,}666.67}$.
- **Stage 2 (Foreign Money Market Investment, $t = 0$ to $t = 1$):** Invest £ 666,666.67 in the UK at 8% $\implies$ Maturity Value $= £\,666{,}666.67 \times 1.08 = £\,\mathbf{720{,}000.00}$.
- **Stage 3 (Forward Exchange Cover, $t = 0$ locked):** Simultaneously sell £ 720,000 forward at $F = \$1.48\,/\text{GBP} \implies$ Forward Inflow $= £\,720{,}000 \times 1.48 = \$\mathbf{1{,}065{,}600.00}$.
- **Stage 4 (Loan Repayment & Profit Extraction, $t = 1$):** Repay US debt: \$1,000,000 × 1.05 = \$**1,050,000.00**. Net riskless profit: \$1,065,600 − \$1,050,000 = \$**15,600.00**.

2. Arbitrage Cash Flow Ledger:

Operational Stage	Cash Flow (USD)	Cash Flow (GBP)
Time $t = 0$ Cash Flows:		
Borrow in US money market (1 year @ 5%)	+\$1,000,000.00	—
Spot conversion to Sterling (@ \$1.50 /GBP)	−\$1,000,000.00	+£ 666,666.67
Invest in UK money market (1 year @ 8%)	—	−£ 666,666.67
Net Cash Flow at Time $t = 0$	**\$0.00**	**£ 0.00**
Time $t = 1$ Cash Flows:		
UK investment maturity proceeds (£ 666,666.67 × 1.08)	—	+£ 720,000.00
Deliver GBP under forward contract (@ \$1.48 /GBP)	+\$1,065,600.00	−£ 720,000.00
Repay US loan principal plus interest (\$1M × 1.05)	−\$1,050,000.00	—
Net Riskless Arbitrage Profit at Time $t = 1$	**+\$15,600.00**	**£ 0.00**

3. Theoretical Parity Benchmark Comparison:
Under theoretical CIP, no-arbitrage forward rate $F^* = S\left(\frac{1+i_{\text{US}}}{1+i_{\text{UK}}}\right) = 1.50\left(\frac{1.05}{1.08}\right) = \$\mathbf{1.4583}\,/\text{GBP}$. Because the actual market quote $F = \$1.4800 > \$1.4583$, the forward pound is overvalued relative to parity (trading at a smaller discount of −1.33% versus −2.78%). This pricing misalignment creates the riskless arbitrage opportunity that yields \$**15,600.00** without initial equity commitment.

Comprehensive Formula Reference Sheet

This section consolidates all mathematical formulas, pricing relationships, valuation models, and operational quotation mechanisms established throughout the course.

Chapter 1: International Finance Overview & Institutions

Balance of Payments Equilibrium Identity

$$\text{BOP} = \text{Current Account} + \text{Capital Account} + \text{Financial Account} + \text{Errors \& Omissions} = 0$$

Where:

- **Current Account:** Net balance on visible merchandise trade, invisible services, primary investment income, and unilateral transfers.
- **Capital Account:** Net cross-border capital transfers, debt forgiveness, and acquisition/disposal of non-produced, non-financial assets.
- **Financial Account:** Net foreign direct investment (FDI), foreign portfolio investment (FPI), commercial loans, and official reserve transactions.
- **Errors & Omissions:** Statistical balancing item ensuring double-entry equality.

SDR Multi-Currency Basket Valuation

$$\text{SDR Value in USD} = \sum_{i=1}^{5} (q_i \times S_i)$$

Where:

- q_i: Fixed units of currency i defined in the 5-year IMF SDR valuation basket.
- S_i: Spot market exchange rate of currency i expressed in US Dollars.
- **Relative Basket Weights:** USD (43.39%), EUR (29.31%), CNY (12.28%), JPY (7.59%), GBP (7.44%).

Chapter 2: The Foreign Exchange Market

Bid-Ask Spread and Percentage Spread

$$\text{Spread} = \text{Ask Price} - \text{Bid Price}$$

$$\text{Percentage Spread } (\%) = \left[\frac{\text{Ask Price} - \text{Bid Price}}{\text{Ask Price}}\right] \times 100$$

Where:

- **Ask Price (Offer):** Exchange rate at which the quoting market maker sells foreign currency.
- **Bid Price:** Exchange rate at which the quoting market maker buys foreign currency.

Covered Interest Rate Parity (Exact Forward Rate)

$$F = S \times \left[\frac{1 + r_h \times \frac{d}{365}}{1 + r_f \times \frac{d}{365}}\right]$$

Where:

- F: Outright forward exchange rate (Home currency per unit of foreign currency).
- S: Spot exchange rate (Home currency per unit of foreign currency).
- r_h: Annualized risk-free interest rate in the domestic (home) country.
- r_f: Annualized risk-free interest rate in the foreign country.
- d: Maturity duration of the forward contract in days.

Linear Forward Approximation & Annualized Premium/Discount

$$F_{\text{linear}} \approx S \times \left[1 + (r_h - r_f) \times \frac{d}{365}\right]$$

$$\text{Annualized Premium / Discount } (\%) = \left[\frac{F - S}{S}\right] \times \frac{365}{d} \times 100$$

Where:

- F: Forward exchange rate; S: Spot exchange rate.
- $r_h - r_f$: Nominal interest rate differential between home and foreign country.
- d: Forward contract duration in days.
- **Condition:** $F > S \implies$ Forward Premium; $F < S \implies$ Forward Discount.

Merchant TT Buying Rate (Clean Remittances)

$$\text{TT Buying Rate} = \text{Interbank Spot Buying Rate (Bid)} - \text{Exchange Margin}$$

Where:

- **Interbank Spot Buying Rate:** Market bid price utilized as base by authorized dealer.
- **Exchange Margin:** Merchant loading percentage deducted to yield the customer buying quote.
- **FEDAI Rounding Convention:** Quoted rate rounded to the nearest multiple of 0.0025.

Merchant Bill Buying Rate (Commercial Trade Bills)

$$\text{Bill Buying Rate} = \text{Forward Buying Rate (for Transit)} - \text{Exchange Margin}$$

$$\text{Forward Buying Rate} = \begin{cases} \text{Spot Bid} - \text{Forward Discount}, & \text{if at Discount} \\ \text{Spot Bid}, & \text{if at Premium} \end{cases}$$

Where:

- **Forward Discount:** Forward discount margin deducted for the normal transit period.
- **Exchange Margin:** Percentage margin deducted from the base buying rate.
- **FEDAI Rounding Convention:** Quoted rate rounded to the nearest multiple of 0.0025.

Recovery of Interest on Commercial Bills Purchased

$$\text{Interest on Bill} = \text{Rupee Amount} \times r \times \left[\frac{\text{NTP} + \text{Usance}}{365}\right]$$

Where:

- **Rupee Amount:** Gross rupee value credited to exporter at the Bill Buying Rate.
- r: Annual commercial bank lending / discount rate charged to customer.
- NTP: Normal Transit Period in days prescribed by FEDAI rules.
- Usance: Commercial credit tenor specified on the bill of exchange in days.

Merchant TT Selling Rate (Outward Remittances)

$$\text{TT Selling Rate} = \text{Interbank Spot Selling Rate (Ask)} + \text{Exchange Margin}$$

Where:

- **Interbank Spot Selling Rate:** Market ask / offer rate at which bank covers in interbank tier.
- **Exchange Margin:** Merchant loading percentage added to base selling rate.
- **FEDAI Rounding Convention:** Quoted rate rounded to the nearest multiple of 0.0025.

Cross-Rate Merchant TT Buying Rate

$$\text{TT Buying Rate}_{\text{FC}} = \frac{\text{Interbank Spot Buying Rate (Bid)}_{\text{USD/INR}} - \text{Exchange Margin}}{\text{Overseas Market Spot Selling Rate (Ask)}_{\text{USD/FC}}}$$

Where:

- Interbank Spot Buying Rate $(\text{Bid})_{\text{USD/INR}}$: Quoting dealer's local base buying price for US Dollars.
- Exchange Margin: Loading percentage deducted from the domestic base rate.
- Overseas Market Spot Selling Rate $(\text{Ask})_{\text{USD/FC}}$: Interbank offer rate for foreign currency in overseas market (Singapore, Tokyo, London).
- Rounding Convention: Rate rounded off to nearest multiple of 0.0025.

Financial Outcome on Foreign Demand Draft Cancellation

$$\begin{aligned}\text{Net Financial Outcome} &= \text{Cancellation Proceeds} - \text{Original Cost} \\ &= \text{FC Amount} \times (\text{TT Buying Rate} - \text{Original Issue Rate})\end{aligned}$$

Where:

- TT Buying Rate: Net cross-rate applied by the bank to repurchase the cancelled draft.
- Original Issue Rate: Exchange rate paid by the customer when the draft was initially issued.
- Condition: Outcome $> 0 \implies$ Customer Realizes Exchange Gain.
- Condition: Outcome $< 0 \implies$ Customer Incurs Exchange Loss.

Forward Delivery Horizon under FEDAI Rule 7

$$\text{Spot Baseline Date} = \text{Trade Booking Date } (T) + 2 \text{ Business Days}$$

$$\text{Fixed Maturity Date} = \text{Spot Baseline Date} + \text{Contract Tenor}$$

$$\text{Option Delivery Period} \leq 1 \text{ Calendar Month}$$

Where:

- T: Transaction booking date.
- Spot Baseline Date: Inception anchor $(T + 2)$ from which forward duration commences.
- Contract Tenor: Duration of forward cover (e.g., 1 month, 2 months, 3 months).
- Option Delivery Period: Flexible window within which customer may deliver foreign exchange.

Forward Merchant Bill Buying Rate with Transit and Usance Tenors

$$\text{Forward Bill Buying Rate} = (S_{\text{Bid}} \pm \text{Swap Points}_M) - [(S_{\text{Bid}} \pm \text{Swap Points}_M) \times m]$$

Where:

- S_{Bid}: Interbank spot buying (bid) base rate for US Dollars.
- Swap Points_M: Monthly forward margin where tenor M = Forward Period + Transit Period + Usance Period.
- Ascending Margins (Premium): Add premium; fractional duration rounded down to lower whole month ($M \leftarrow \lfloor M \rfloor$).
- Descending Margins (Discount): Deduct discount; fractional duration rounded up to higher whole month ($M \leftarrow \lceil M \rceil$).
- m: Statutory exchange margin percentage (e.g., 0.10%).
- Rounding Rule: Final merchant quote rounded to nearest multiple of 0.0025.

FEDAI Option Forward Delivery Margin Selection Rule

Bank Purchases FX (Merchant Sells):

$$\begin{cases} \text{FC at Premium} \implies \text{Earliest Option Date (Lowest Premium Paid)} \\ \text{FC at Discount} \implies \text{Latest Option Date (Highest Discount Deducted)} \end{cases}$$

Bank Sells FX (Merchant Buys):

$$\begin{cases} \text{FC at Premium} \implies \text{Latest Option Date (Highest Premium Charged)} \\ \text{FC at Discount} \implies \text{Earliest Option Date (Lowest Discount Allowed)} \end{cases}$$

Where:

- Option Delivery Window: Contractual execution window not exceeding 1 calendar month.
- Principle of Quotation: Quoting dealer prices assuming merchant exercises option on date least favorable to bank.
- Ascending Points: Indicates foreign currency forward premium.
- Descending Points: Indicates foreign currency forward discount.

Cross-Rate Forward Merchant Conversion Identity

$$F_{\text{Buying (FC)}} = \frac{F_{\text{Domestic Interbank Buying (USD/INR)}}}{F_{\text{Overseas Interbank Selling (USD/FC)}}}$$

$$F_{\text{Selling (FC)}} = \frac{F_{\text{Domestic Interbank Selling (USD/INR)}}}{F_{\text{Overseas Interbank Buying (USD/FC)}}}$$

Where:

- $F_{\text{Domestic Interbank Buying (USD/INR)}}$: Mumbai forward dollar buying rate (Spot Bid $\pm$ forward margin less margin).
- $F_{\text{Overseas Interbank Selling (USD/FC)}}$: Overseas market forward dollar selling rate (Spot Ask $\pm$ overseas forward margin).
- $F_{\text{Domestic Interbank Selling (USD/INR)}}$: Mumbai forward dollar selling rate (Spot Ask $\pm$ forward margin plus margin).
- $F_{\text{Overseas Interbank Buying (USD/FC)}}$: Overseas market forward dollar buying rate (Spot Bid $\pm$ overseas forward margin).
- Rounding Convention: Final synthesized cross-rate rounded to nearest multiple of 0.0025.

Relative Purchasing Power Parity (PPP) & Multi-Period Compounding

$$S_t = S_0 \left(\frac{1 + i_L}{1 + i_F} \right)^t$$

$$S_t = S_0 \prod_{k=1}^{t} \left(\frac{1 + i_{L,k}}{1 + i_{F,k}} \right)$$

Where:

- S_0: Initial spot exchange rate expressed as domestic currency per unit of foreign currency.
- S_t: Expected spot exchange rate at future time horizon t.
- i_L: Expected inflation rate in the local (domestic/terms) country.
- i_F: Expected inflation rate in the foreign (base) country.
- $i_{L,k}, i_{F,k}$: Term-structure inflation rates prevailing during specific annual period k.
- t: Time horizon in years.

The Fisher Effect & Nominal Rate Decomposition

$$1 + r_{\text{nominal}} = (1 + r_{\text{real}})(1 + i_{\text{inflation}})$$

$$r_{\text{nominal}} \approx r_{\text{real}} + i_{\text{inflation}}$$

Where:

- r_{nominal}: Nominal market interest rate quoted on debt contracts.
- r_{real}: Real rate of return required by investors in the absence of inflation.
- $i_{\text{inflation}}$: Anticipated or expected inflation rate over the contractual period.
- International Link: Real rates equalize across open capital markets ($r_{\text{real},d} \approx r_{\text{real},f}$), implying nominal interest differentials reflect relative inflation expectations ($i_d - i_f \approx \pi_d - \pi_f$).

Covered Interest Rate Parity (CIP) & Forward Margin Condition

$$F = S \times \left(\frac{1 + i_d}{1 + i_f} \right)$$

$$\frac{F - S}{S} \approx i_d - i_f$$

Where:

- F: Contractual forward exchange rate quoted outright.
- S: Current spot exchange rate (direct quote: domestic currency per unit of foreign currency).
- i_d: Nominal interest rate prevailing in the domestic money market for tenor matching F.
- i_f: Nominal interest rate prevailing in the foreign money market for tenor matching F.
- $\frac{F-S}{S}$: Annualized percentage forward premium (> 0) or discount (< 0) on the foreign currency.
- Arbitrage Condition: When $F \neq S \times \frac{1+i_d}{1+i_f}$, covered interest arbitrage generates riskless profit until market equilibrium is restored.